UNIVERSITI TEKNIKAL MALAYSIA MELAKA

FAKULTI TEKNOLOGI MAKLUMAT DAN KOMUNIKASI

W O R K S H O P 1

R E P O R T

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Matric Number : B032310752

Program : 2BITI (S1G1)

Project Title : CINEMA BOOKING TICKET SYSTEM

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Supervisor Signature:

EXECUTIVE SUMMARY

The Cinema Ticket Booking System was developed to ease the always cumbersome and time consuming process of booking movie tickets. Traditional methods include buying tickets at the cinema or through third party operators, which may bring about inconveniences like long queues, limited availability of preferred seats, or delayed confirmations of bookings. With the integration of technology in daily life, an efficient and user friendly system was conceptualized to address these challenges.

The scope of the project work is to design an educational level cinema ticket booking system that will not only provide the basic facilities of movie ticket booking but will also include features like meal selection, loyalty points, and discounts for students, among others. It is designed keeping in mind the needs of frictionless interaction by users and comprehensive management functions for the administrators in today's digital first world. The project aspires to make cinema going pleasant and hassle free for one and all by promoting clarity, access, and an improved user experience.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

The Cinema Ticket Booking System is a timely creation to help address concerns from movie enthusiasts and cinema administrators alike in conventional methodologies of ticket booking. With digitalization growing by the minute, this calls for a platform that will not only be smooth and effective but also user friendly in line with today's pace.

The project aims at developing an integrated platform where users can view movie information, choose seats, book tickets, and make their payments without any hassle. Simultaneously, administrators will have a facility for managing bookings, movies, payments, and user data accurately and efficiently. It enhances the user experience by including features such as student discounts, meal combos, and loyalty points that also increase user engagement and retention.

1.2 Problem Statement

In many cinemas, managing ticket bookings, payments, and seat allocations is often done manually or using basic tools like spreadsheets. This approach can lead to errors, such as double bookings, incorrect payment entries, or mismanagement of seat availability. It can also be slow to find information about movie schedules, bookings, or customer data when needed. These inefficiencies make running a cinema challenging and sometimes result in customer dissatisfaction, lost revenue, or unnecessary delays in service delivery.

1.3 Objective (s) of the project

1. **To facilitate a seamless and user-friendly cinema booking experience**

    i. The system enables users to easily browse movies, select showtimes, choose seats, and complete payments efficiently.

    ii. Provides a structured and intuitive interface for users and admins, ensuring a smooth and hassle free booking process.

2. **To enhance booking management and operational efficiency**

    i. The system reduces manual errors by automating the management of user accounts, bookings, and payment processing.

    ii. Ensures real time seat availability tracking, preventing double bookings and improving cinema space utilization.

3. **To provide valuable insights for cinema administrators**

    i. Generates detailed reports on sales trends, movie popularity, and customer preferences to support data driven decision-making.

    ii. Helps admins optimize scheduling, improve promotions, and enhance the overall movie going experience for customers.

1.4 Scope

1.4.1 Modules to be developed

1. **User Management Module**:

    i. Allows admins to manage user accounts, including adding, updating, viewing, and deleting users.

    ii. Facilitates student verification and approval for discounts.

    iii. Enables users to register, log in, and manage their profiles securely.

2. **Booking Management Module**:

    i. Enables users to browse movies, select seats, and book tickets.

    ii. Admins can view, add, and update booking records as necessary.

    iii. Ensures real time seat availability to prevent double bookings.

3. **Movie Management Module**:

    i. Admins can add, update, delete, and view movie details, including showtimes, genres, and ratings.

    ii. Maintains an updated list of currently available movies for users to browse.

4. **Payment Management Module**:

    i. Processes payments with secure card validation and discount applications (e.g., student discounts and loyalty points).

    ii. Allows admins to add, update, view, and delete payment records.

    iii. Generates receipts for completed transactions.

5. **Meal Management Module**:

    i. Users can select meals or combos while booking tickets.

    ii. Admins can manage meal options, including adding, updating, and deleting items from the menu.

6. **Report Module**:

i. Admins can generate detailed reports on booking trends, revenue, movie popularity, seat utilization, and monthly breakdowns.

ii. Includes tools for analysing data, visualizing trends, and identifying opportunities for operational improvement.

iii. Provides interactive options to view year-wise and month-wise reports with detailed booking data for each selected period.

1.4.2 Target User

i. **Users:** Those who want to book movie tickets with ease and comfort from anywhere and prefer to have the freedom to choose movies, seats and even food, without having to visit the cinema in person.

ii. **Student users:** Students looking for special discounts when booking tickets, get a 40% discount.

iii. **Administrators:** Who want tools to oversee cinema operations, analyze data, and generate reports that include insights into decision making, user accounts, bookings, payments, movies and meals.

1.5 Project Significance

i. Improved User Experience:

Simplifies movie ticket booking with the facility of browsing and booking seats conveniently from anywhere.

More personalized features loyalty points, meal combos, and student discounts make the experience even more engaging and rewarding.

In depth real time information regarding seat availability and movie schedules will reduce frustration and pain among users, thereby saving them time.

ii. Operational Efficiency for Administrators:

Smoothest out administrative procedures, from user account management down to booking records and bill payments, through one central system. Manually entered mistakes regarding bookings and payments are reduced, with a positive effect on accuracy. The data analysis and reporting features allow better decision making and operational planning.

iii. Increased Revenue and Customer Retention:

Loyalty points and targeted discounts encourage repeat bookings and enhance customer retention.

Having combo options generates another revenue stream at a cinema, complementing profit margins.

Having combo options enables consumer satisfaction, which gives greater support to word of mouth for the facility to see other movies.

iv. Scalability and Adaptability:

Modular system is designed for the future: The system is compatible with a mobile app platform and multilingual language system;

Suitable for small, medium sized and even larger cinemas.

v. The Educational Value :

This provides comprehensive learning for the developers involved in the project, starting from database management to user interface design, secure payment handling, and report generation.

This is a good example of how programming and system design principles are applied to real life projects, which is very useful for academic and personal growth.

1.6 Gantt Chart of Project Activities

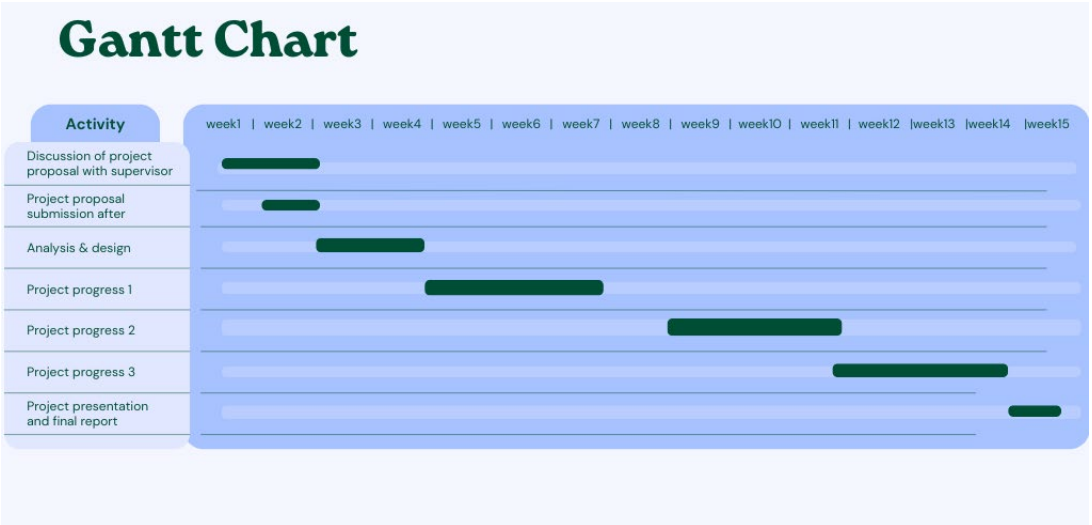

Figure 1.1: Gantt Chart

CHAPTER 2: ANALYSIS OF PROBLEM

2.1 Problem Decomposition Description

Sub Problem	Description	Steps to Solve
1. User Management	Manage user registration, login, and profile updates, including student verification for discounts.	- Create a database table to store user details, including student status and admin privileges. - Implement registration and login functions with secure authentication. - Design an admin feature for verifying student status based on matriculation numbers.
2. Booking Management	Allow users to book tickets, select seats, and view booking history while preventing double bookings.	- Design a booking database to store movie, seat, and user details. - Develop a seat selection module with real time availability updates. - Implement validation to prevent duplicate bookings for the same seat.
3. Movie Management	Maintain a catalog of movies, including showtimes, genres, and ratings.	- Create a movie database to store details like title, genre, rating, duration, and showtimes. - Develop admin features to add, update, or delete movies and their schedules. - Enable users to search and browse movies by various criteria.
4. Payment Management	Handle secure payments with options for discounts.	- Create a payment database to store transaction details. - Implement functions for secure payment processing, including card validation. - Apply discounts based on student status or loyalty points.
5. Report Generation	Provide admins with tools to generate reports on revenue, booking trends, and seat utilization.	- Identify key metrics (e.g., revenue, movie popularity, booking frequency). -Allows admins to filter reports by year and month to view specific booking and revenue details. - Validates input ranges to ensure accurate data retrieval and report generation.

2.2 Structured Chart

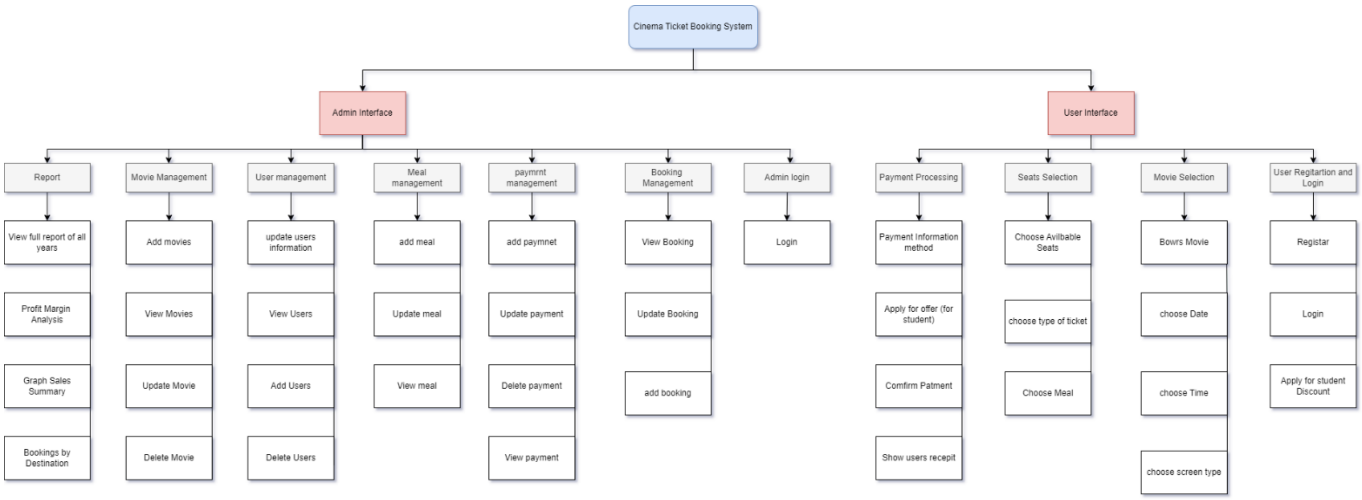

Figure 2.1: Structured chart

CHAPTER 3: DESIGN

3.1 Flowchart

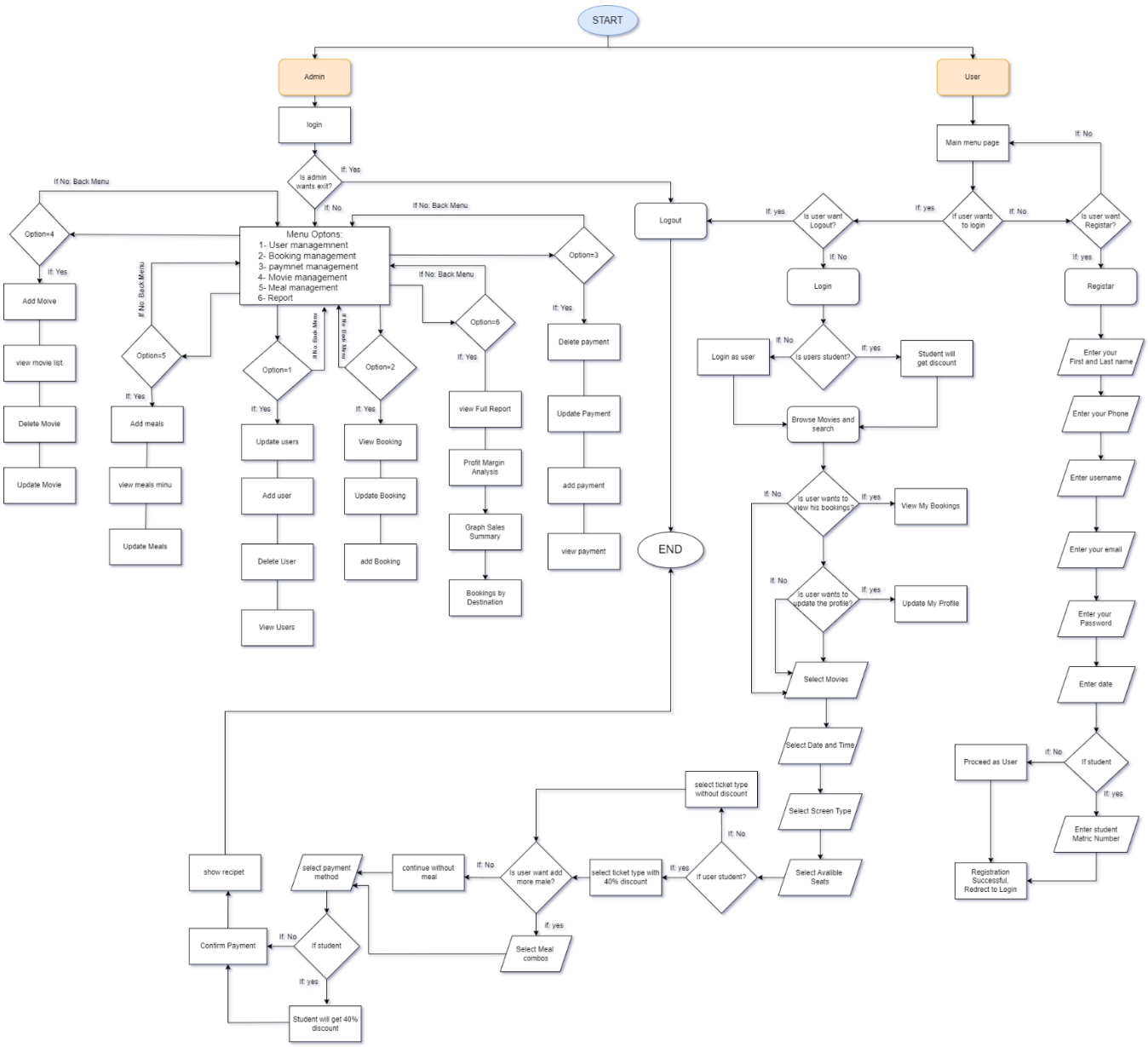

Figure 3.1: Workflow and functional module overview

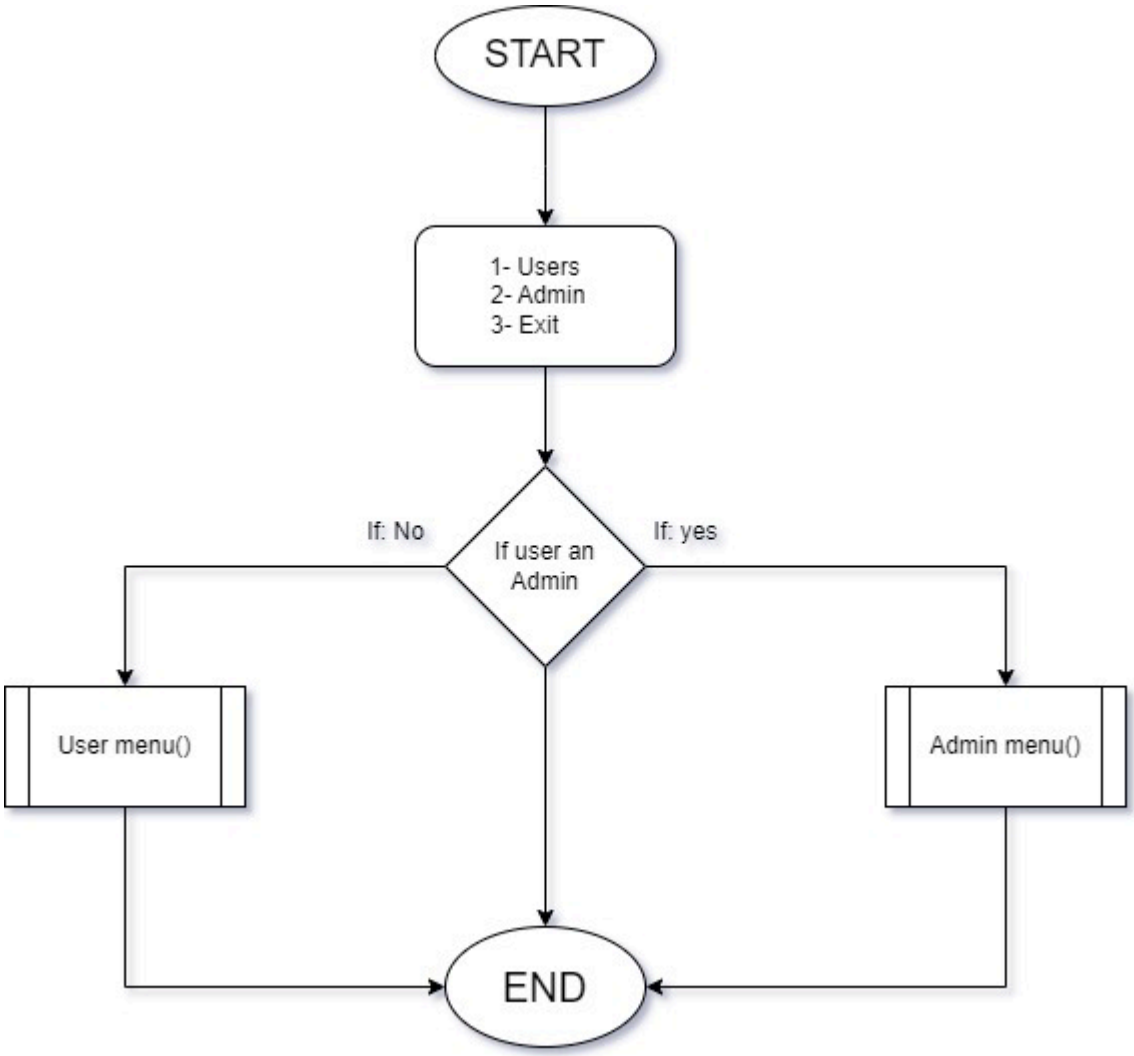

Figure 3.2: Register and login as user & student, and login as admin

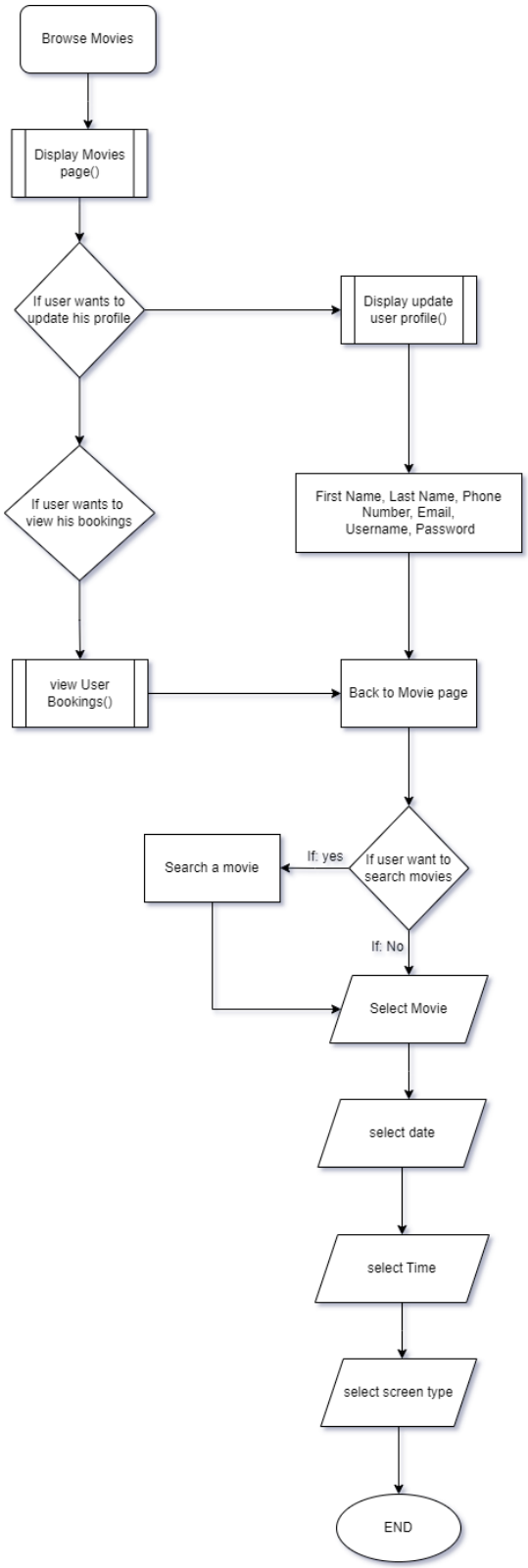

Figure 3.3: Brows movie and select booking date & time, select screen type

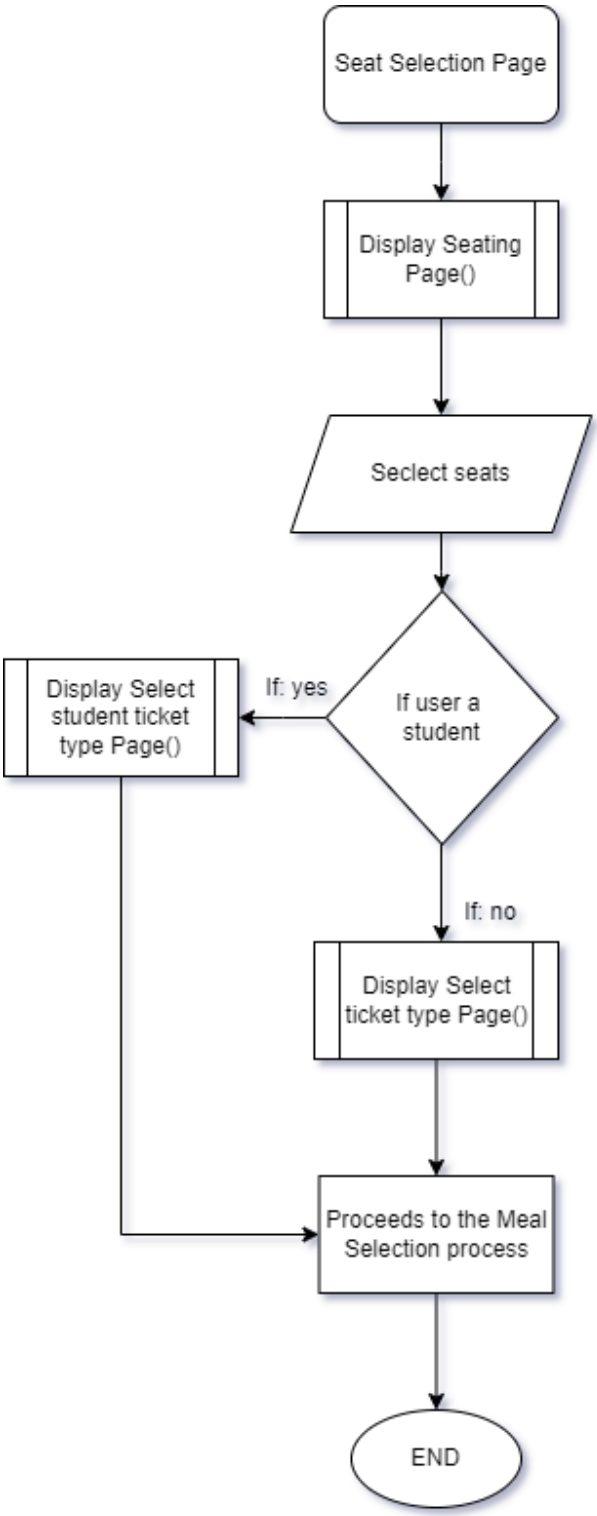

Figure 3.4: Select seats and type of ticket for user & student, select meal

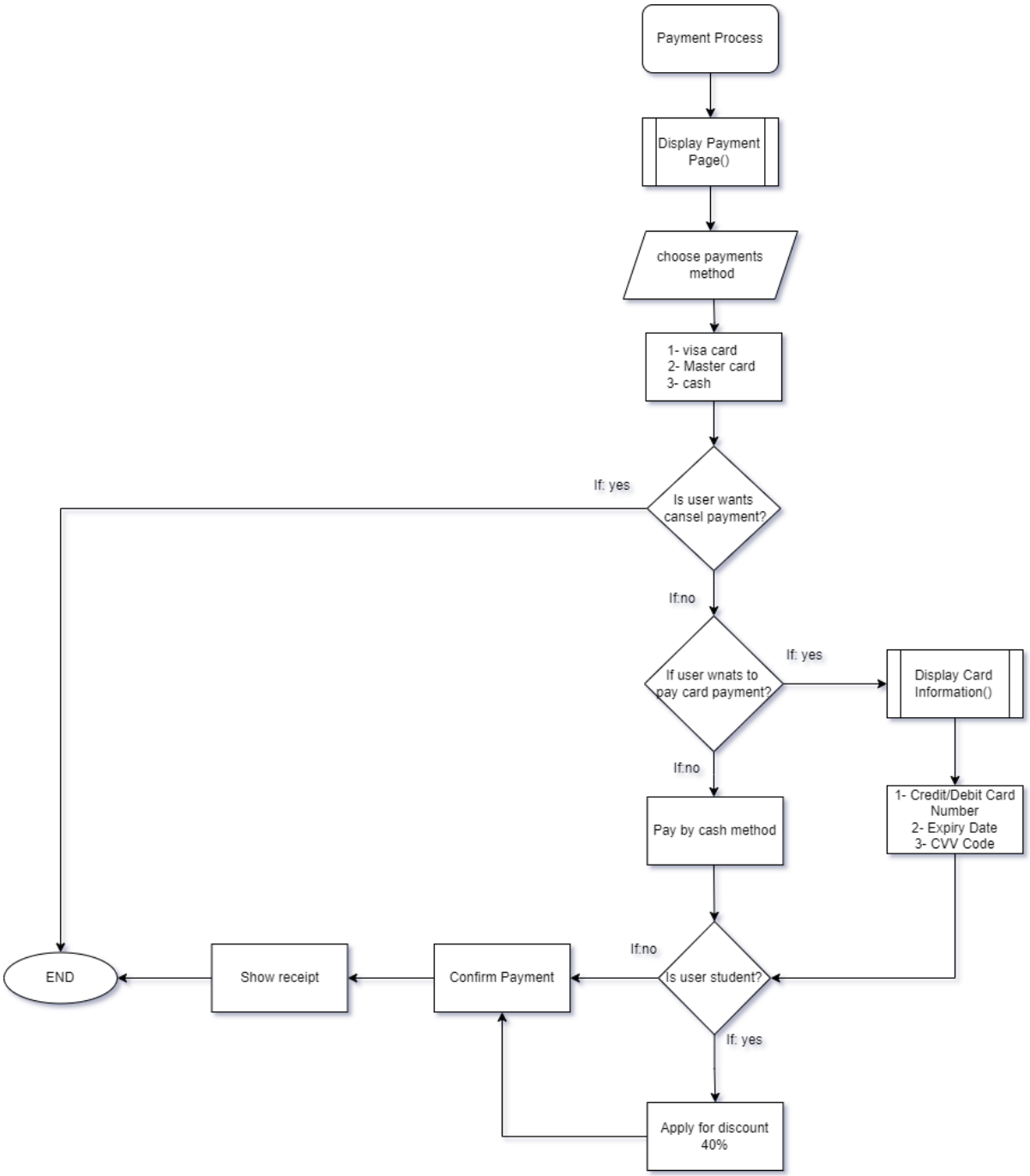

Figure 3.5: Payment Process flow

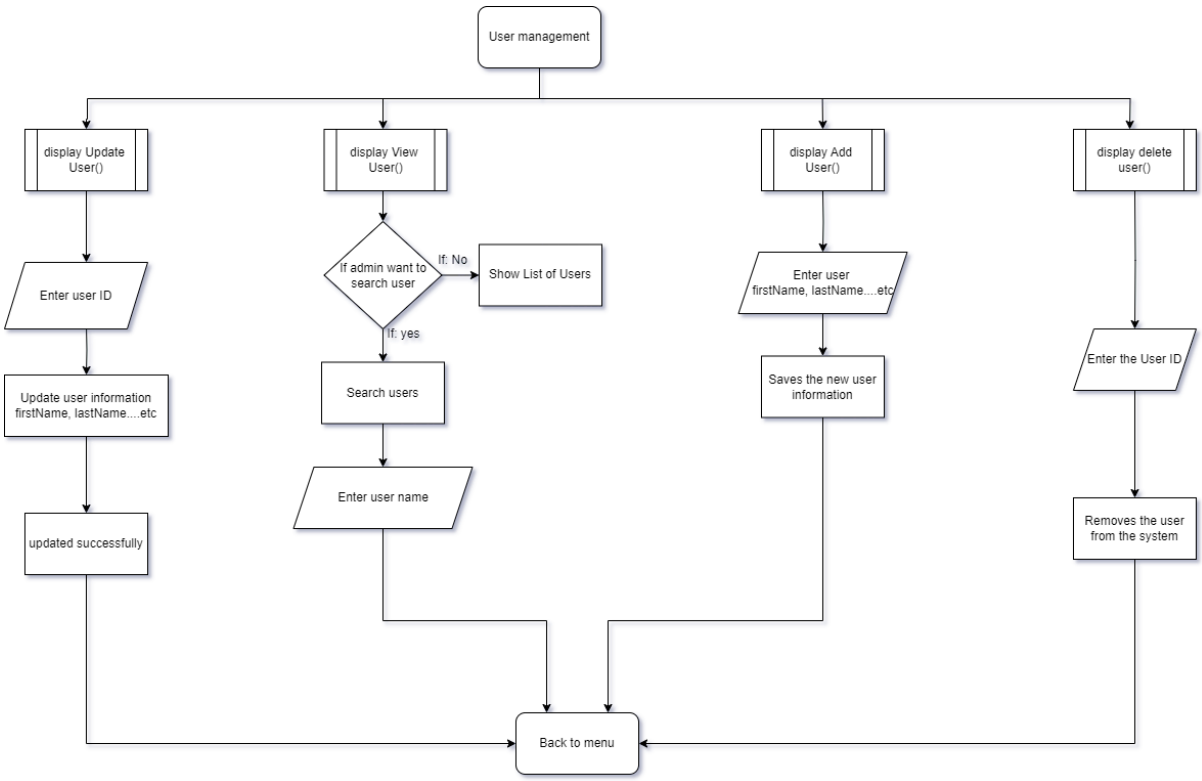

Figure 3.6: Users management

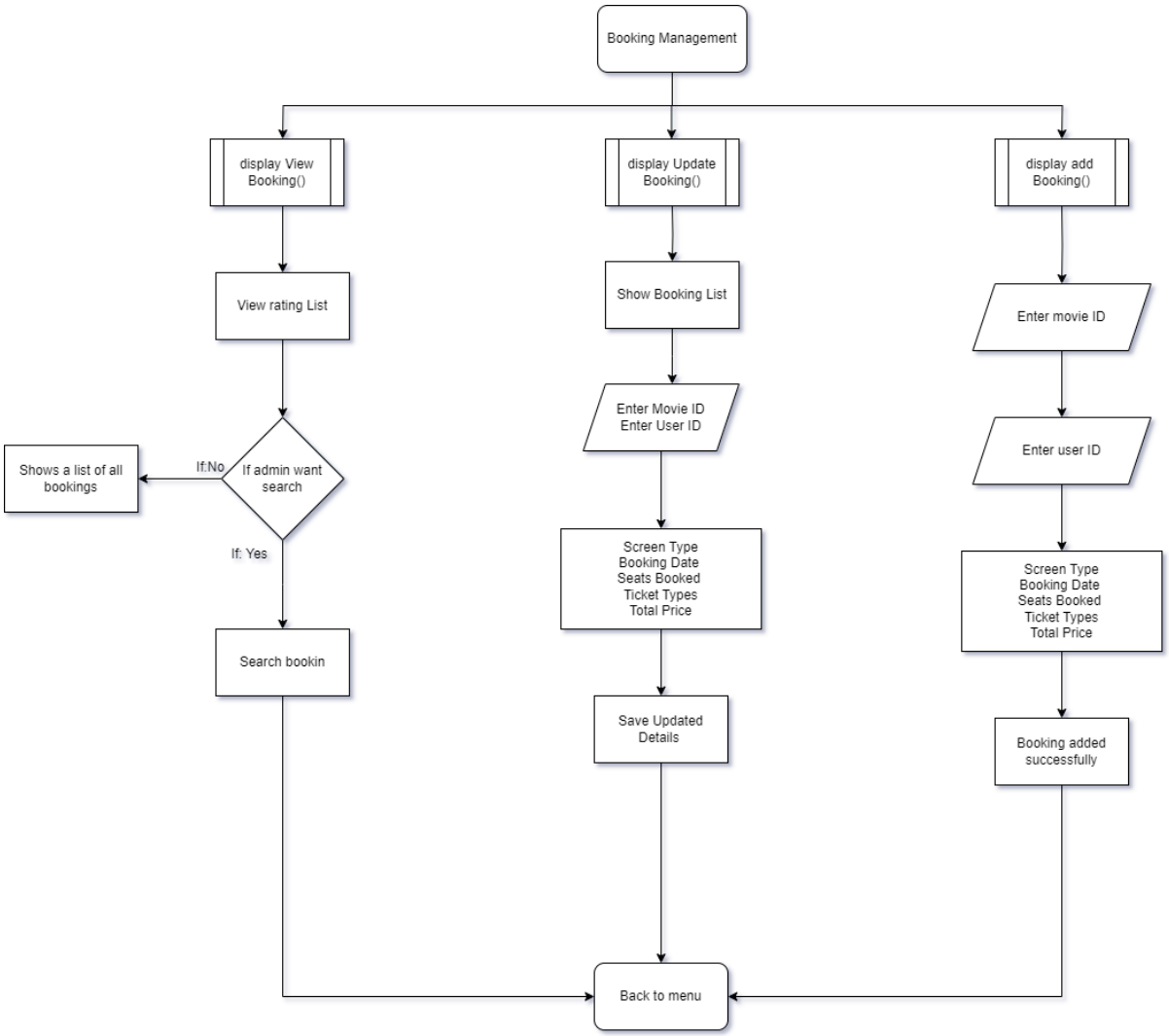

Figure 3.7: Booking management

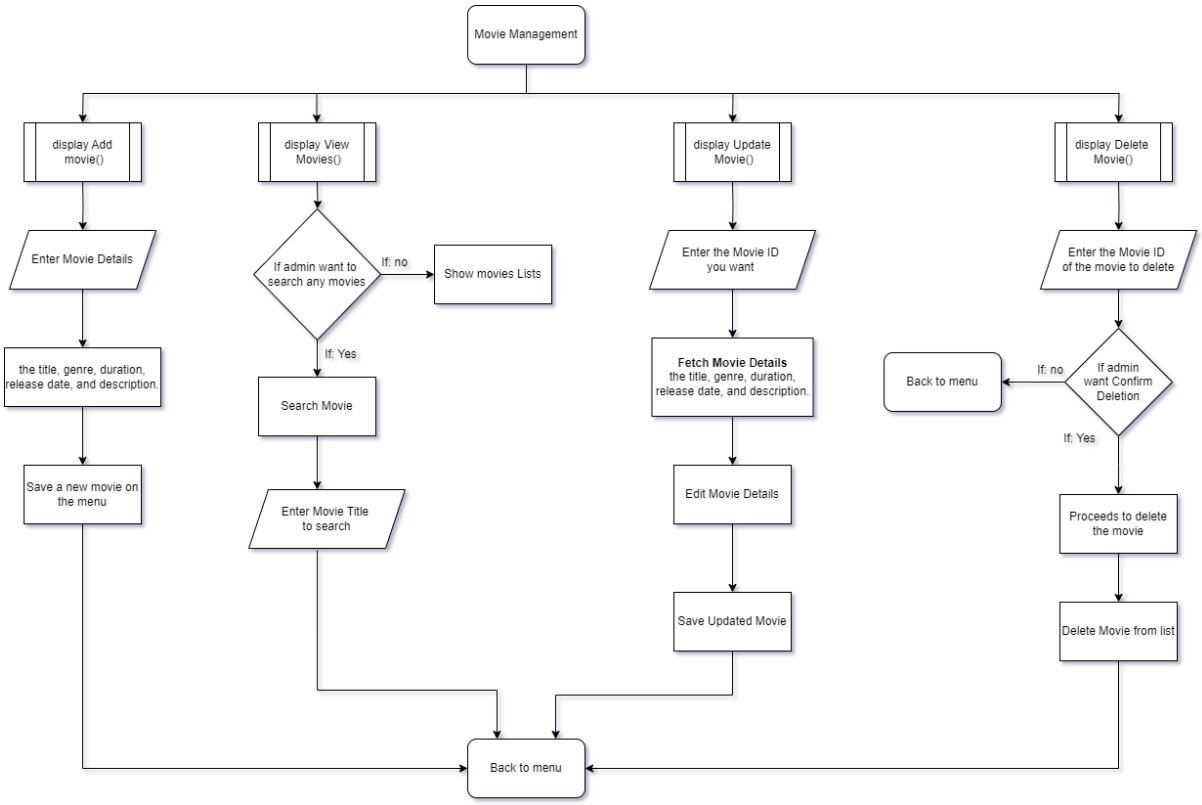

Figure 3.8: Movie management

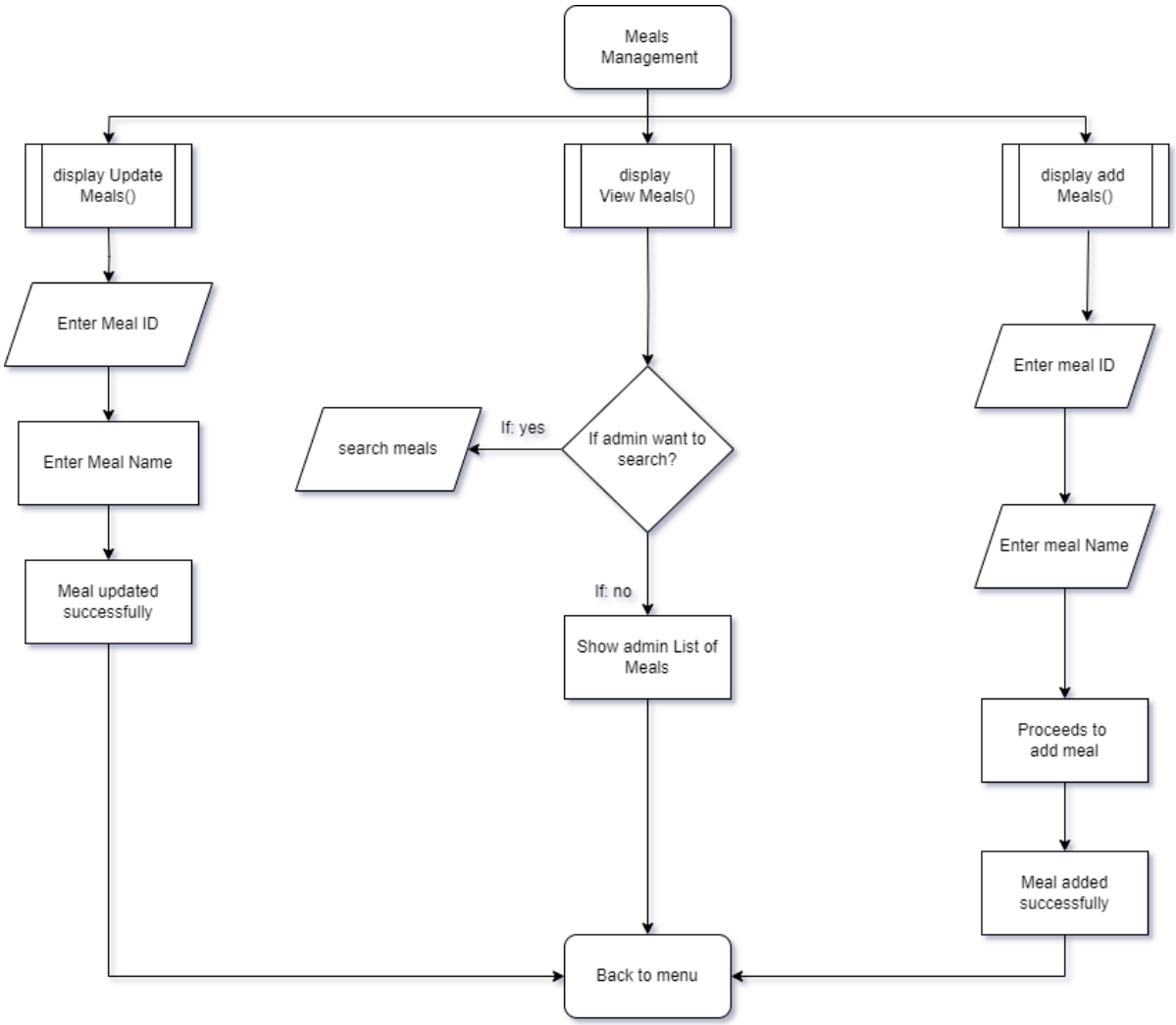

Figure 3.9: Meals management

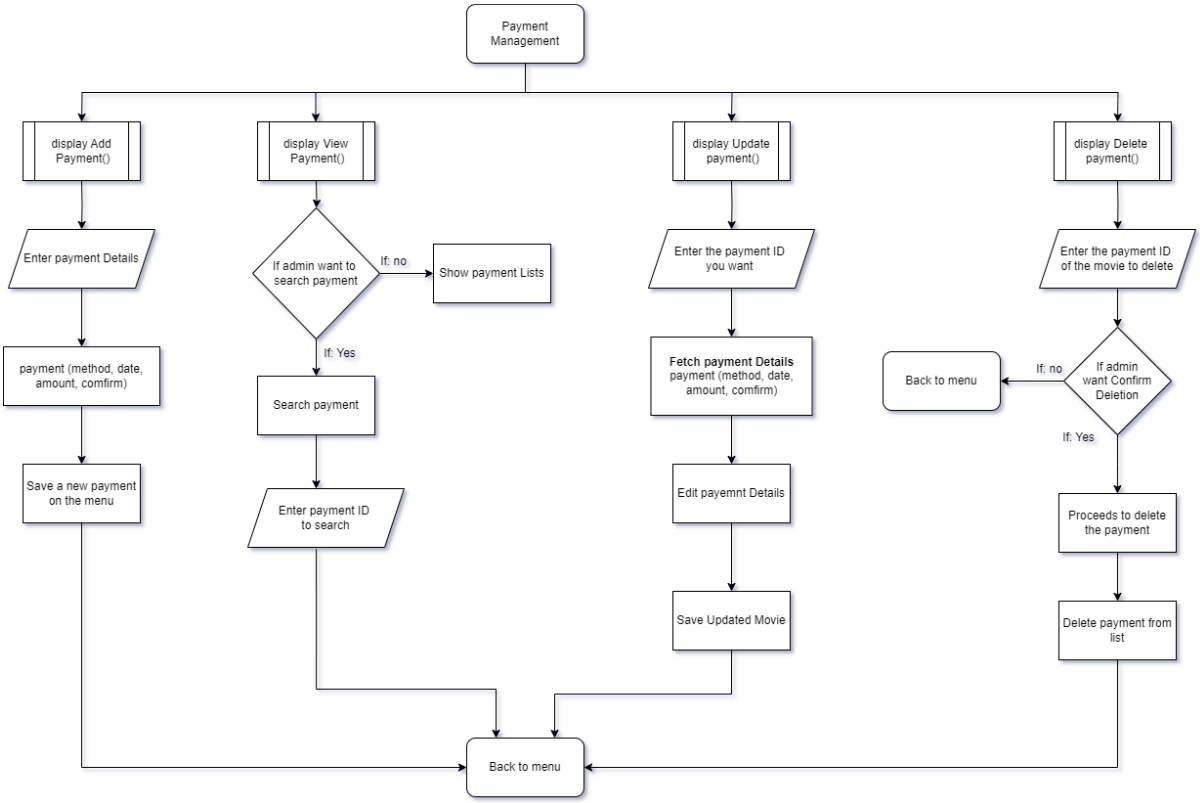

Figure 3.10: Payment management

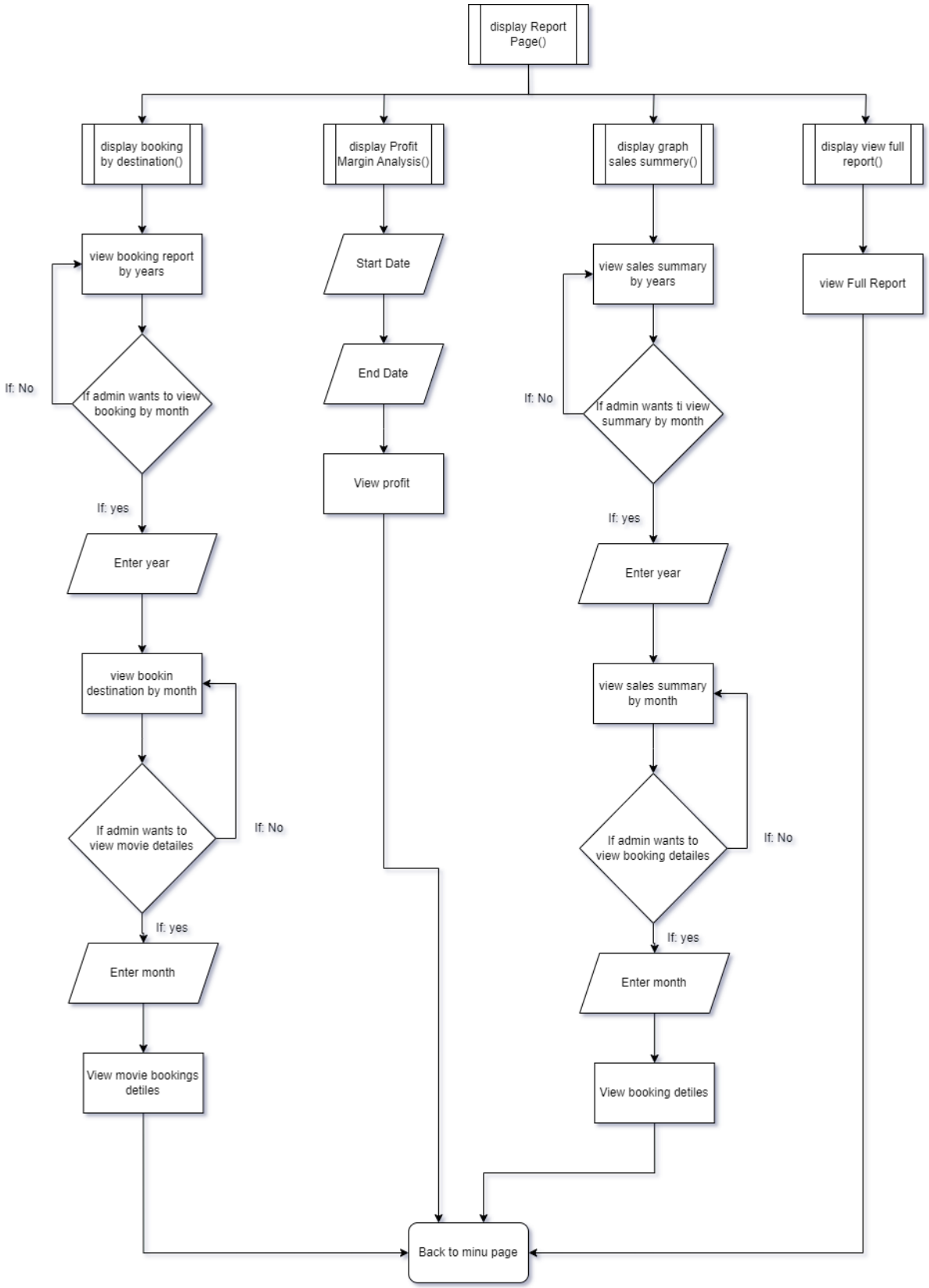

Figure 3.11: Report process

3.2 ERD

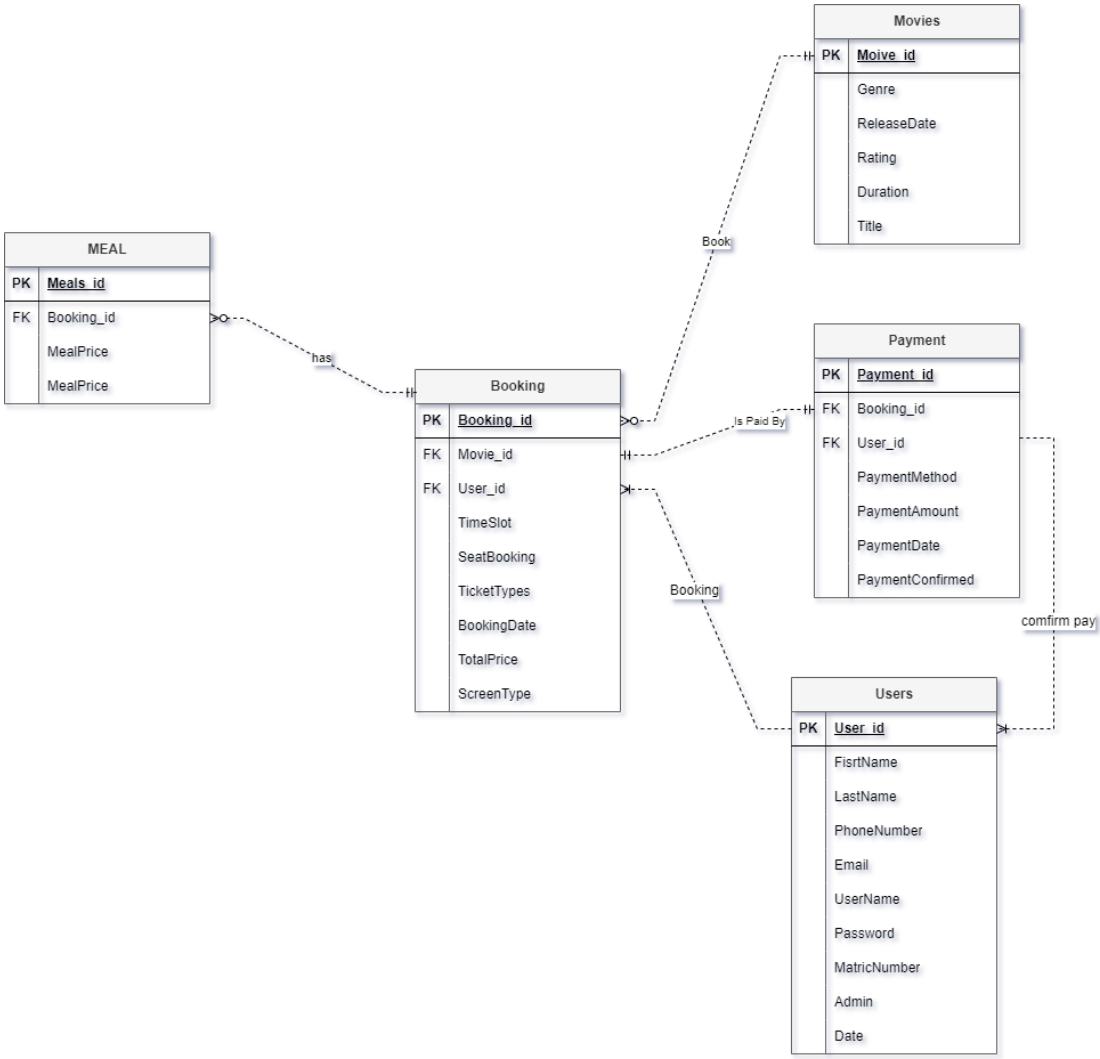

Figure 3.12: ERD

3.3 Data Dictionary

Booking:

Table 3.13: Booking table

Field	Type	Null	Key	Default	Extra
BookingID	int(11)	NO	PRI	*NULL*	auto_increment
Movie_id	int(11)	YES	MUL	*NULL*	
User_id	int(11)	YES	MUL	*NULL*	
ScreenType	varchar(50)	YES		*NULL*	
BookingDate	date	YES		*NULL*	
TimeSlot	varchar(20)	YES		*NULL*	
SeatsBooked	varchar(255)	YES		*NULL*	
TicketTypes	varchar(255)	YES		*NULL*	
TotalPrice	double	NO		0	

Users:

Table 3.14: Users table

Field	Type	Null	Key	Default	Extra
User_id	int(11)	NO	PRI	*NULL*	auto_increment
FirstName	varchar(255)	YES		*NULL*	
LastName	varchar(255)	YES		*NULL*	
PhoneNumber	varchar(15)	YES	MUL	*NULL*	
Email	varchar(255)	YES	UNI	*NULL*	
UserName	varchar(255)	YES	UNI	*NULL*	
Password	varchar(255)	YES	MUL	*NULL*	
Admin	tinyint(1)	YES		0	
Date	date	YES		*NULL*	
MatricNumber	varchar(20)	YES		*NULL*	
IsStudent	tinyint(1)	YES		0	

Meals:

Table: 3.15: Meals table

Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
MealID	int(11)			No	*None*		AUTO_INCREMENT	Change Drop More
MealName	varchar(255)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
MealPrice	decimal(10,2)			Yes	*NULL*			Change Drop More
Booking_id	int(11)			Yes	*NULL*			Change Drop More

Movies:

Table 3.16: Movies table

Field	Type	Null	Key	Default	Extra
Movie_id	int(11)	NO	PRI	*NULL*	auto_increment
Title	varchar(255)	YES		*NULL*	
Genre	varchar(100)	YES		*NULL*	
ReleaseDate	date	YES		*NULL*	
Rating	decimal(3,1)	YES		*NULL*	
Duration	varchar(10)	YES		*NULL*	

Payment:

Table 3.17: Payment table

Field	Type	Null	Key	Default	Extra
Payment_id	int(11)	NO	PRI	*NULL*	auto_increment
User_id	int(11)	YES	MUL	*NULL*	
Booking_id	int(11)	YES	MUL	*NULL*	
PaymentMethod	varchar(20)	YES		*NULL*	
PaymentAmount	decimal(10,2)	YES		*NULL*	
PaymentDate	datetime	YES		current_timestamp()	
PaymentConfirmed	tinyint(1)	YES		0	

3.4 Interface Design

```
*****************************************
*       CINEMA TICKET BOOKING SYSTEM    *
*****************************************

1. REGISTER
2. LOGIN
3. EXIT

Choose an option: |
```

Figure 3.18: Main menu page

```
*****************************************
*                REGISTER               *
*****************************************

Note: Enter '0' at any input prompt to Exit from this page.

    Enter First Name: Alwaleed
    Enter Last Name: Hassan
    Enter Username: Welly
    Enter Phone Number: 60014908386
    Enter Email: welley@gmail.com
    Enter Password: **
    Confirm Password: **
    Enter Date (YYYY-MM-DD): 2003-09-09
    Are you a student? (Y/N): y
    Enter your Matric Number: B032310752

    Registration successful! Returning to the main menu...
```

Figure 3.19: Register page

```
*****************************************
*                LOGIN                  *
*****************************************

Enter Email or Username: welly
Enter Password: **

Logged in as User. Redirecting to Movie Page...
You are logged in as a student. Enjoy a 40% discount on tickets!
```

Figure 3.20: Login as student page

```
                    ************************************************************************
                    *                          MOVIE PAGE                                  *
                    ************************************************************************

Top 9 Recommended Movies:
+-------+-----------------------------------------+-----------------+--------+----------+--------------+
| ID    | Title                                   | Genre           | Rating | Duration | Release Date |
+-------+-----------------------------------------+-----------------+--------+----------+--------------+
| 42    | forzen                                  | animtion        | 10.0   | 2:34     | 2023-09-09   |
| 15    | Avengers: Secret Wars                   | Action          | 9.3    | 2:50     | 2025-11-08   |
| 13    | John Wick: Chapter 5                    | Action          | 9.2    | 2:45     | 2025-03-15   |
| 39    | Star Wars: Rogue Squadron               | Science         | 9.2    | 2:35     | 2025-12-18   |
| 8     | Dune: Part Two                          | Science         | 9.1    | 2:40     | 2024-11-03   |
| 29    | Mad Max: The Wasteland                  | Adventure       | 9.1    | 2:25     | 2025-05-15   |
| 2     | Spider Man                              | Animation       | 9.0    | 2:20     | 2023-06-02   |
| 16    | The Batman: Part II                     | Action          | 9.0    | 2:55     | 2025-03-04   |
| 28    | Deadpool 3                              | Comedy          | 8.9    | 2:14     | 2024-05-03   |
+-------+-----------------------------------------+-----------------+--------+----------+--------------+

Options:
1. Select a Movie by ID
2. View and Search Movies
3. Update My Profile
4. View My Bookings
5. Go Back to Main Menu
Enter your choice:
```

Figure 3.21: Movie page

```
                    ************************************************************************
                    *                          MY BOOKINGS                                 *
                    ************************************************************************

+---------+-----------------------------------------+--------------+----------+----------------------+--------------+
| Booking | Movie Title                             | Date         | Time     | Seats                | Total Price  |
+---------+-----------------------------------------+--------------+----------+----------------------+--------------+
| 258     | Spider Man                              | 2025-01-16   | 10:00 AM | E5                   | 35.70        |
+---------+-----------------------------------------+--------------+----------+----------------------+--------------+
```

Figure 3.22: View user & student page

```
************************************************************************
*                          UPDATE PROFILE                              *
************************************************************************
Options:
1. Update First Name
2. Update Last Name
3. Update Phone Number
4. Update Email
5. Update Username
6. Update Password
7. Back to Profile Page
Enter your choice:
```

Figure 3.23: Update profile page

```
*********************************************************************
*                            SEARCH PAGE                            *
*********************************************************************

Available Movies:
+-------+----------------------------------------+-----------------+--------+----------+--------------+
| ID    | Title                                  | Genre           | Rating | Duration | Release Date |
+-------+----------------------------------------+-----------------+--------+----------+--------------+
| 42    | forzen                                 | animtion        | 10.0   | 2:34     | 2023-09-09   |
| 15    | Avengers: Secret Wars                  | Action          | 9.3    | 2:50     | 2025-11-08   |
| 13    | John Wick: Chapter 5                   | Action          | 9.2    | 2:45     | 2025-03-15   |
| 39    | Star Wars: Rogue Squadron              | Science         | 9.2    | 2:35     | 2025-12-18   |
| 8     | Dune: Part Two                         | Science         | 9.1    | 2:40     | 2024-11-03   |
| 29    | Mad Max: The Wasteland                 | Adventure       | 9.1    | 2:25     | 2025-05-15   |
| 2     | Spider Man                             | Animation       | 9.0    | 2:20     | 2023-06-02   |
| 16    | The Batman: Part II                    | Action          | 9.0    | 2:55     | 2025-03-04   |
| 28    | Deadpool 3                             | Comedy          | 8.9    | 2:14     | 2024-05-03   |
| 38    | Doctor Strange                         | Fantasy         | 8.9    | 2:30     | 2025-06-20   |
| 4     | Oppenheimer                            | Drama           | 8.8    | 3:00     | 2023-07-21   |
| 30    | Zootopia 2                             | Animation       | 8.8    | 1:56     | 2025-07-10   |
| 25    | Fantastic Four                         | Science         | 8.7    | 2:20     | 2024-11-08   |
| 40    | Toy Story 5                            | Animation       | 8.7    | 1:55     | 2025-11-15   |
| 14    | Frozen III                             | Animation       | 8.5    | 1:55     | 2025-06-10   |
| 37    | The Lion King: Mufasa                  | Animation       | 8.4    | 1:57     | 2024-07-12   |
| 7     | Guardians Galaxy                       | Action          | 8.3    | 2:30     | 2023-05-05   |
| 23    | The Hunger Games                       | Drama           | 8.2    | 2:37     | 2024-11-17   |
| 35    | Kung Fu Panda 4                        | Animation       | 8.2    | 1:40     | 2024-03-08   |
| 1     | Avatar                                 | drama           | 8.1    | 3:12     | 2023-01-15   |
| 26    | Inside Out 2                           | Animation       | 8.1    | 1:40     | 2024-06-14   |
| 5     | Mission: Impossible                    | Action          | 8.0    | 2:56     | 2023-07-14   |
| 12    | Wonka                                  | Adventure       | 8.0    | 2:10     | 2024-12-13   |
| 24    | Blade                                  | Action          | 8.0    | 2:18     | 2024-09-06   |
| 36    | The Matrix Resurrections               | Science         | 8.0    | 2:45     | 2024-12-15   |
| 11    | Aquaman                                | Action          | 7.9    | 2:30     | 2024-12-25   |
| 21    | Elemental                              | Animation       | 7.9    | 1:42     | 2023-06-16   |
| 20    | Super Mario Bros                       | Animation       | 7.8    | 1:32     | 2023-04-07   |
| 32    | Creed III                              | Drama           | 7.8    | 1:56     | 2023-03-03   |
| 10    | Transformers                           | Action          | 7.5    | 2:15     | 2024-06-09   |
| 18    | Indiana Jones                          | Adventure       | 7.5    | 2:22     | 2023-06-30   |
| 27    | The Equalizer 3                        | Thriller        | 7.5    | 1:49     | 2023-09-01   |
| 6     | Barbie                                 | Comedy          | 7.4    | 1:54     | 2023-07-21   |
| 34    | Black Panther                          | Action          | 7.4    | 2:41     | 2023-01-20   |
| 17    | Shazam! Fury                           | Fantasy         | 7.3    | 2:10     | 2023-03-17   |
| 3     | The Flash                              | Action          | 7.2    | 2:24     | 2023-07-10   |
| 19    | Fast X                                 | Action          | 7.2    | 2:20     | 2023-05-19   |
| 22    | Blue Beetle                            | Action          | 7.1    | 2:07     | 2023-08-18   |
| 31    | The Little Mermaid                     | Fantasy         | 7.1    | 2:15     | 2023-05-26   |
| 9     | The Marvels                            | Action          | 7.0    | 2:10     | 2024-02-14   |
| 44    | Hallo                                  | action          | 7.0    | 2:09     | 2025-12-12   |
+-------+----------------------------------------+-----------------+--------+----------+--------------+

Options:
1. Enter the Movie ID to select a movie
2. Search for a movie by keyword
3. Go back to the main movie page
Enter your choice: |
```

Figure 3.24: Search & movie menu

```
*********************************************************************
*                        SEARCH BY KEYWORD                          *
*********************************************************************

Enter a keyword to search for a movie: sp

Movies Matching Your Search:
+-------+----------------------------------------+-----------------+--------+----------+--------------+
| ID    | Title                                  | Genre           | Rating | Duration | Release Date |
+-------+----------------------------------------+-----------------+--------+----------+--------------+
| 2     | Spider Man                             | Animation       | 9.0    | 2:20     | 2023-06-02   |
+-------+----------------------------------------+-----------------+--------+----------+--------------+

Enter the Movie ID to select a movie or 0 to go back: |
```

Figure 3.25: Search movie

```
*********************************************************
*                      SELECT A DAY                     *
*********************************************************

          +------------+----------------+
          |   Option   |      Day       |
          +------------+----------------+
          | 1          | 2025-01-16     |
          | 2          | 2025-01-17     |
          | 3          | 2025-01-18     |
          | 4          | 2025-01-19     |
          | 5          | 2025-01-20     |
          | 6          | 2025-01-21     |
          | 7          | 2025-01-22     |
          | 8          | 2025-01-23     |
          | 9          | 2025-01-24     |
          | 10         | 2025-01-25     |
          | 11         | 2025-01-26     |
          +------------+----------------+

Choose the day you want (or 0 to go back): |
```

Figure 3.26: Select date

```
*********************************************************
*                    SELECT A TIME SLOT                 *
*********************************************************

          +------------+-------------------+
          |   Option   |     Time Slot     |
          +------------+-------------------+
          | 1          | 10:00 AM          |
          | 2          | 01:00 PM          |
          | 3          | 04:00 PM          |
          | 4          | 07:00 PM          |
          | 5          | 10:00 PM          |
          +------------+-------------------+

Choose a time slot (or 0 to go back): |
```

Figure 3.27: Select time slot

```
*********************************************************
*                    SELECT SCREEN TYPE                 *
*********************************************************

          +------------+-------------------+
          |   Option   |    Screen Type    |
          +------------+-------------------+
          | 1          | Standard          |
          | 2          | IMAX              |
          | 3          | 3D                |
          | 4          | VIP               |
          +------------+-------------------+

Choose the screen type (or 0 to go back): |
```

Figure 3.28: Select screen type

```
                                                                    ********************************************************************
                                                                    *                          SEAT SELECTION                          *
                                                                    ********************************************************************

                                                                            Movie: Spider Man | Genre: Animation | Rating: 9.0 | Duration: 2:20
                                                                            Date: 2025-01-16 | Time: 10:00 AM | Screen Type: Standard

                                                                            Seats:
                                                                             A1    A2    A3    A4    A5    A6    A7    A8
                                                                             B1    B2    B3    B4    B5    B6    B7    B8
                                                                             C1    C2    C3    C4    C5    C6    C7    C8
                                                                             D1    D2    D3    D4    D5    D6    D7    D8
                                                                             E1    E2    E3    E4    E5    E6    E7    E8
                                                                             F1    F2    F3    F4    F5    F6    F7    F8
                                                                             G1    G2    G3    G4    G5    G6    G7    G8
                                                                             H1    H2    H3    H4    H5    H6    H7    H8

Instructions:
1. Enter seat names to select (e.g., A1).
2. Type 'done' to finalize your selection.
3. Type 'remove' to remove a selected seat.
4. Type '0' to return to the previous page.
                                                                    Your choice: E5
```

Figure 3.29: Seat selection page

```
*********************************************************************
*                      TICKET SELECTION PAGE                        *
*********************************************************************

Selected Seats:
- D3
- D4

+-------------------+---------------------+
| Ticket Type       | Price per Ticket    |
+-------------------+---------------------+
| Individual        | RM 22.00            |
| Kids              | RM 15.00            |
| Senior            | RM 18.00            |
+-------------------+---------------------+

For Seat D3, select a ticket type (1/2/3): 1
For Seat D4, select a ticket type (1/2/3): 2

Total Price: RM 37.00
Do you want to proceed with the booking?
1. Confirm Tickets
2. Go Back to Seat Selection
Enter your choice: 1

Tickets successfully selected! Proceeding to meal selection...
```

Figure 3.30: Ticket selection for student

```
*********************************************************************
*                        STUDENT TICKET SELECTION                   *
*********************************************************************

You have selected the following seats:
+---------------------+---------------------+
|        Seat         |      Price (RM)     |
+---------------------+---------------------+
| E5                  | 15.00               |
+---------------------+---------------------+
Total Price: RM 15.00

1. Proceed to Meal Selection
2. Go Back to Seat Selection
Enter your choice:
```

Figure 3.31: Ticket selection for users

```
*********************************************************
*                        MEAL SELECTION                 *
*********************************************************

          Available Meals:
          +------------+------------------+-------------+
          |   Option   |       Meal       |   Price     |
          +------------+------------------+-------------+
          | 1          | Popcorn (S)      | 5.00        |
          | 2          | Popcorn (M)      | 7.50        |
          | 3          | Popcorn (L)      | 10.00       |
          | 4          | Soft Drink (S)   | 3.00        |
          | 5          | Soft Drink (M)   | 5.00        |
          | 6          | Soft Drink (L)   | 6.50        |
          | 7          | cocola           | 10.50       |
          | 8          | vanta            | 14.00       |
          | 9          | Nachos with Cheese | 8.00       |
          | 10         | Hot Dog          | 6.50        |
          | 11         | Chicken Nuggets  | 7.00        |
          | 12         | Fries (Small)    | 4.00        |
          | 13         | Fries (Large)    | 6.00        |
          | 14         | Ice Cream Cup    | 4.50        |
          | 15         | Water Bottle     | 2.00        |
          | 16         | chocata          | 21.00       |
          | 17         | Frise            | 9.00        |
          +------------+------------------+-------------+

          Selected Meals:
          +------------------+-------------+
          |       Meal       |   Price     |
          +------------------+-------------+
          | Popcorn (M)      | 7.50        |
          | Popcorn (S)      | 5.00        |
          +------------------+-------------+
          Total Price: RM 12.50

          Instructions:
          1. Enter the meal number to select a meal.
          2. Type 'remove' to remove a selected meal by its number.
          3. Type 'done' to proceed to payment.
          4. Type '0' to return to the previous page.
          Your choice: 3
          Added Popcorn (L) to your selection.
```

Figure 3.32: Meals selection page

```
*********************************************************
*                      PAYMENT SUMMARY                  *
*********************************************************
+-----------------+-----------------+-----------------+-----------------+
| Ticket Type     | Number of Tickets | Price per Ticket | Total Price (RM) |
+-----------------+-----------------+-----------------+-----------------+
| Student         | 1               | 15.00           | 15.00           |
+-----------------+-----------------+-----------------+-----------------+

*********************************************************
*                        MEALS                          *
*********************************************************
+-------------------------+-------------------+
| Meal Name               | Price (RM)        |
+-------------------------+-------------------+
| Popcorn (M)             |              7.50 |
| Popcorn (S)             |              5.00 |
| Popcorn (L)             |             10.00 |
+-------------------------+-------------------+
Student Discount Applied (40% Off): RM 15.00
Final Amount to Pay: RM 22.50

Choose Payment Method:
1. Visa
2. MasterCard
3. Cash
4. Cancel Payment
Enter your choice: |
```

```
*********************************************************
*                      PAYMENT SUMMARY                  *
*********************************************************
+-----------------+-----------------+-----------------+-----------------+
| Ticket Type     | Number of Tickets | Price per Ticket | Total Price (RM) |
+-----------------+-----------------+-----------------+-----------------+
| Individual      | 1               | 22.00           | 22.00           |
| Kids            | 1               | 15.00           | 15.00           |
+-----------------+-----------------+-----------------+-----------------+

*********************************************************
*                        MEALS                          *
*********************************************************
+-------------------------+-------------------+
| Meal Name               | Price (RM)        |
+-------------------------+-------------------+
| Popcorn (S)             |              5.00 |
+-------------------------+-------------------+
Final Amount to Pay: RM 42.00

Choose Payment Method:
1. Visa
2. MasterCard
3. Cash
4. Cancel Payment
Enter your choice: 3
```

Figure 3.33: Payment summary page for student

Figure 3.34: Payment summary page for user

```
Choose Payment Method:
1. Visa
2. MasterCard
3. Cash
4. Cancel Payment
Enter your choice: 3

Total to pay: RM 49.00
Enter payment amount: RM 40
You still need to pay: RM 9.00

Total to pay: RM 49.00
Enter payment amount: RM 9
```

Figure 3.35: Cash method

```
Card Number (16 digits): 9288736522128973
Expiry Date (MM/YY): 09/28
CVV (3 digits): 223

1. Confirm Payment
2. Go Back to Payment Page
Enter your choice: 1

Payment successful! Redirecting to receipt...
```

Figure 3.36: Card method

```
*********************************************************************
*                              RECEIPT                              *
*********************************************************************
| User Name: Alwaleed Hassan                                        |
| Movie:     Spider Man                                             |
| Showtime:  2025-01-16                                             |
| Receipt Date: 2025-01-16                                          |
| Receipt Time: 12:13:05                                            |
*********************************************************************
*                            TICKET DETAILS                         *
*********************************************************************
+-------------------+-------------------+
| Ticket Type       | Quantity          |
+-------------------+-------------------+
| Student           | 1                 |
+-------------------+-------------------+
*********************************************************************
*                            MEAL DETAILS                           *
*********************************************************************
+-------------------------+-------------------+
| Meal Name               | Meal Price (RM)   |
+-------------------------+-------------------+
| Popcorn (M)             |              7.50 |
| Popcorn (S)             |              5.00 |
| Popcorn (L)             |             10.00 |
+-------------------------+-------------------+
*********************************************************************
*                           PRICE SUMMARY                           *
*********************************************************************
| Total Price Before Discount: RM 22.50                             |
| Discounted Price:           RM 13.50                              |
*********************************************************************
*                  THANK YOU, Alwaleed Hassan! ENJOY YOUR MOVIE!    *
*********************************************************************
```

Figure 3.37: Student receipt

```
*********************************************************************
*                              RECEIPT                              *
*********************************************************************
| User Name: Asjed tomaro                                          |
| Movie:     Spider Man                                            |
| Showtime:  2025-01-16                                            |
| Receipt Date: 2025-01-16                                         |
| Receipt Time: 13:22:16                                           |
*********************************************************************
*                            TICKET DETAILS                         *
*********************************************************************
+-------------------+-------------------+
| Ticket Type       | Quantity          |
+-------------------+-------------------+
| Individual        | 1                 |
| Kids              | 1                 |
+-------------------+-------------------+
*********************************************************************
*                            MEAL DETAILS                           *
*********************************************************************
+-------------------------+-------------------+
| Meal Name               | Meal Price (RM)   |
+-------------------------+-------------------+
| Popcorn (S)             |              5.00 |
+-------------------------+-------------------+
*********************************************************************
*                            PRICE SUMMARY                          *
*********************************************************************
| Final Price:              RM 42.00                               |
*********************************************************************
*                  THANK YOU, Asjed tomaro! ENJOY YOUR MOVIE!
*********************************************************************
```

Figure 3.38: User receipt

```
*****************************************
*                  LOGIN                *
*****************************************

Enter Email or Username: alwaleed24
Enter Password: **********

Logged in as Admin. Redirecting to Admin Page...
Press any key to continue . . .
```

Figure 3.39: Login as admin

```
*****************************************************************
*                            ADMIN PAGE                         *
*****************************************************************

1. Manage Users
2. Manage Payments
3. Manage Meals
4. Manage Movies
5. Manage Bookings
6. View Reports
7. Exit Admin Page

Enter your choice:
```

Figure 3.40: Admin page

~USERS MANAGEMENT~

```
*****************************************************************
*                          MANAGE USERS                         *
*****************************************************************

1. View All Users
2. Add New User
3. Update User Information
4. Delete User
5. Return to Admin Page

Enter your choice: |
```

Figure 3.41: Users management page

```
*****************************************************************
*                        VIEW ALL USERS                         *
*****************************************************************

Options:
1. View All Users
2. Search User
3. Return to Manage Users page
Enter your choice: 2

Enter Name to search: y

+-------+----------------+----------------+-------------------+-------------------+-----------------+-------+----------------+------------+
| UserID| First Name     | Last Name      | Email             | Username          | Phone Number    | Admin | MatricNumber   | IsStudent  |
+-------+----------------+----------------+-------------------+-------------------+-----------------+-------+----------------+------------+
|   145 |         Hassan |          Elyas |     alw@gmail.com |          hassan22 |    09128373632 |    No |   b0323107622 |        Yes |
|   148 |           Lily |         Walker |    lily@gmail.com |             lilyw |     0123456778 |    No |           N/A |         No |
|   153 |          Daisy |         Taylor |   daisy@gmail.com |            daisyt |     0123456786 |    No |        M123456 |        Yes |
|   173 |         Hannah |         Taylor |  hannah@gmail.com |           hannaht |     0112233448 |    No |           N/A |         No |
+-------+----------------+----------------+-------------------+-------------------+-----------------+-------+----------------+------------+
```

```
*****************************************************************
*                        VIEW ALL USERS                         *
*****************************************************************

+-------+----------------+----------------+-------------------+-------------------+-----------------+-------+----------------+------------+
| UserID| First Name     | Last Name      | Email             | Username          | Phone Number    | Admin | MatricNumber   | IsStudent  |
+-------+----------------+----------------+-------------------+-------------------+-----------------+-------+----------------+------------+
| 2     | Asjed          | tomaro         | aj@gmail.com      | asjed             | 0908112121      | No    |                | No         |
| 145   | Hassan         | Elyas          | alw@gmail.com     | hassan22          | 09128373632     | No    | b0323107622    | Yes        |
| 146   | Abdulah        | khaled         | abdu22@gmail.com  | abdu220           | 1234566789      | No    | B032310752     | Yes        |
| 147   | Ahmed          | naseer         | hamody@gmail.com  | hamody12          | 0981276665      | No    | b21334566      | Yes        |
| 148   | Lily           | Walker         | lily@gmail.com    | lilyw             | 0123456778      | No    | N/A            | No         |
| 149   | Nina           | Evans          | nina@gmail.com    | ninae             | 0123456776      | No    | N/A            | No         |
| 150   | Alice          | Johnson        | alice@gmail.com   | alicej            | 0123456789      | No    | N/A            | No         |
| 151   | Bob            | Smith          | bob@gmail.com     | bobsmith          | 0123456788      | No    | N/A            | No         |
| 152   | Charlie        | Brown          | charlie@gmail.com | charlieb          | 0123456787      | No    | N/A            | No         |
| 153   | Daisy          | Taylor         | daisy@gmail.com   | daisyt            | 0123456786      | No    | M123456        | Yes        |
+-------+----------------+----------------+-------------------+-------------------+-----------------+-------+----------------+------------+

Navigation:
2. Next Page
3. Return to Manage Users page
Enter your choice: |
```

Figure 3.42: View all students and users & search

```
*****************************************************************
*                          ADD NEW USER                         *
*****************************************************************
Note: Enter '0' at any input prompt to return to the previous page.

Enter First Name: ml
Enter Last Name: jh
Enter Email: ll@gmail.com
Enter Username: til
Enter Phone Number: 12354667892
Enter Password: mkk
Confirm Password: mkk
Enter Date (YYYY-MM-DD): 2004-08-22
Is Admin (1 for Yes, 0 for No): 0

User added successfully!

Would you like to add another user? (y/n): |
```

Figure 3.43: Add new user

```
*************************************************************
*                 UPDATE USER INFORMATION                   *
*************************************************************

+--------+-----------------+-----------------+-----------------------+-----------------+-------+-----------+-----------------+
| UserID| First Name      | Last Name       | Email                 | Username        | Phone Number    | Admin | IsStudent | Matric Number   |
+--------+-----------------+-----------------+-----------------------+-----------------+-------+-----------+-----------------+
|      2 |           Asjed |          tomaro |       aj@gmail.com    |           asjed |     0908112121 |    No |        No |                 |
|    145 |          Hassan |            Elyas |      alw@gmail.com    |        hassan22 |  09128373632 |    No |       Yes |      b0323107622 |
|    146 |         Abdulah |          khaled |   abdu22@gmail.com    |         abdu220 |     1234566789 |    No |       Yes |       B032310752 |
|    147 |           Ahmed |          naseer |   hamody@gmail.com    |        hamody12 |     0981276665 |    No |       Yes |        b21334566 |
|    148 |            Lily |          Walker |     lily@gmail.com    |            lilyw |     0123456778 |    No |        No |              N/A |
|    149 |            Nina |           Evans |     nina@gmail.com    |           ninae |     0123456776 |    No |        No |              N/A |
|    150 |           Alice |         Johnson |    alice@gmail.com    |           alicej |     0123456789 |    No |        No |              N/A |
|    151 |             Bob |           Smith |      bob@gmail.com    |        bobsmith |     0123456788 |    No |        No |              N/A |
|    152 |         Charlie |           Brown |   charlie@gmail.com   |         charlieb |     0123456787 |    No |        No |              N/A |
|    153 |           Daisy |          Taylor |    daisy@gmail.com    |          daisyt |     0123456786 |    No |       Yes |          M123456 |
|    154 |            Evan |        Williams |     evan@gmail.com    |           evanw |     0123456785 |    No |        No |              N/A |
|    155 |           Fiona |           Clark |     fiona@gmail.com   |           fionac |     0123456784 |    No |        No |              N/A |
|    156 |          George |          Miller |   george@gmail.com    |          georgem |     0123456783 |    No |       Yes |          M654321 |
|    157 |          Hannah |           Davis |      nah@gmail.com    |         hannahd |     0123456782 |    No |        No |              N/A |
|    158 |             Ian |           Moore |      ian@gmail.com    |            ianm |     0123456781 |    No |        No |              N/A |
|    159 |           Julia |        Anderson |    julia@gmail.com    |           juliaa |     0123456780 |    No |       Yes |         M987654 |
|    160 |           Kevin |          Harris |    kevin@gmail.com    |          kevinh |     0123456779 |    No |        No |              N/A |
|    162 |            Mike |           Scott |     mike@gmail.com    |           mikesc |     0123456777 |    No |       Yes |         M123789 |
|    164 |          Oscar |          Wilson |    oscar@gmail.com    |           oscarw |     0123456775 |    No |        No |              N/A |
|    165 |           Paula |            Reed |     paula@gmail.com   |           paular |     0123456774 |    No |       Yes |          M456123 |
|    166 |       Alwaleed |          Hassan |    welly@gmail.com    |       almaleed24 |   60149408286 |   Yes |        No |              N/A |
|    167 |        Brianna |         Johnson |   brianna@gmail.com   |         briannaj |     0112233442 |    No |        No |              N/A |
|    168 |    Christopher |           Brown |    chris@gmail.com    |           chrisb |     0112233443 |    No |       Yes |        UTEM12345 |
|    169 |          Diana |           Davis |    diana@gmail.com    |           dianad |     0112233444 |    No |        No |              N/A |
|    170 |          Ethan |          Miller |    ethan@gmail.com    |           ethanm |     0112233445 |    No |       Yes |        UTEM67890 |
|    171 |          Fiona |          Wilson |    fiofo@gmail.com    |           fionaw |     0112233446 |    No |        No |              N/A |
|    172 |         George |           Moore |    gerge@gmail.com    |           gergem |     0112233447 |    No |       Yes |        UTEM23456 |
|    173 |         Hannah |          Taylor |   hannah@gmail.com    |          hannaht |     0112233448 |    No |        No |              N/A |
|    174 |          Isaac |        Anderson |    isaac@gmail.com    |           isaaca |     0112233449 |    No |       Yes |        UTEM78901 |
|    175 |          Julia |          Thomas |     juli@gmail.com    |           juliat |     0112233450 |    No |        No |              N/A |
|    188 |         Albert |           Smith |   albert@gmail.com    |          alberts |     0112233441 |    No |        No |              N/A |
|    189 |            mna |             kla |      lala@gmail.com   |             lala |   12121212121 |    No |       Yes |         b918811 |
|    190 |       Alwaleed |          Hassan |   welley@gmail.com    |           Welly |   60014908386 |    No |       Yes |       B032310752 |
+--------+-----------------+-----------------+-----------------------+-----------------+-------+-----------+-----------------+

Enter the User ID to update or 0 to return: 2

Options:
1. Update First Name
2. Update Last Name
3. Update Phone Number
4. Update Email
5. Update Username
6. Update Password
7. Update Admin (1 for Yes, 0 for No)
8. Update Is Student (1 for Yes, 0 for No)
9. Update Matric Number
10. Return to Manage Users

Enter your choice: |
```

Figure 3.44: Update users page

```
*************************************************************
*                        DELETE USER                        *
*************************************************************

+--------+-----------------+-----------------+-----------------------+-----------------+-----------------+-------+
| UserID| First Name      | Last Name       | Email                 | Username        | Phone Number    | Admin |
+--------+-----------------+-----------------+-----------------------+-----------------+-----------------+-------+
|      2 |           Asjed |          tomaro |       aj@gmail.com    |           asjed |     0908112121 |    No |
|    145 |          Hassan |            Elyas |      alw@gmail.com    |        hassan22 |  09128373632 |    No |
|    146 |         Abdulah |          khaled |   abdu22@gmail.com    |         abdu220 |     1234566789 |    No |
|    147 |           Ahmed |          naseer |   hamody@gmail.com    |        hamody12 |     0981276665 |    No |
|    148 |            Lily |          Walker |     lily@gmail.com    |            lilyw |     0123456778 |    No |
|    149 |            Nina |           Evans |     nina@gmail.com    |           ninae |     0123456776 |    No |
|    150 |           Alice |         Johnson |    alice@gmail.com    |           alicej |     0123456789 |    No |
|    151 |             Bob |           Smith |      bob@gmail.com    |        bobsmith |     0123456788 |    No |
|    152 |         Charlie |           Brown |   charlie@gmail.com   |         charlieb |     0123456787 |    No |
|    153 |           Daisy |          Taylor |    daisy@gmail.com    |          daisyt |     0123456786 |    No |
|    154 |            Evan |        Williams |     evan@gmail.com    |           evanw |     0123456785 |    No |
|    155 |           Fiona |           Clark |     fiona@gmail.com   |           fionac |     0123456784 |    No |
|    156 |          George |          Miller |   george@gmail.com    |          georgem |     0123456783 |    No |
|    157 |          Hannah |           Davis |      nah@gmail.com    |         hannahd |     0123456782 |    No |
|    158 |             Ian |           Moore |      ian@gmail.com    |            ianm |     0123456781 |    No |
|    159 |           Julia |        Anderson |    julia@gmail.com    |           juliaa |     0123456780 |    No |
|    160 |           Kevin |          Harris |    kevin@gmail.com    |          kevinh |     0123456779 |    No |
|    162 |            Mike |           Scott |     mike@gmail.com    |           mikesc |     0123456777 |    No |
|    164 |          Oscar |          Wilson |    oscar@gmail.com    |           oscarw |     0123456775 |    No |
|    165 |           Paula |            Reed |     paula@gmail.com   |           paular |     0123456774 |    No |
|    166 |       Alwaleed |          Hassan |    welly@gmail.com    |       almaleed24 |   60149408286 |   Yes |
|    167 |        Brianna |         Johnson |   brianna@gmail.com   |         briannaj |     0112233442 |    No |
|    168 |    Christopher |           Brown |    chris@gmail.com    |           chrisb |     0112233443 |    No |
|    169 |          Diana |           Davis |    diana@gmail.com    |           dianad |     0112233444 |    No |
|    170 |          Ethan |          Miller |    ethan@gmail.com    |           ethanm |     0112233445 |    No |
|    171 |          Fiona |          Wilson |    fiofo@gmail.com    |           fionaw |     0112233446 |    No |
|    172 |         George |           Moore |    gerge@gmail.com    |           gergem |     0112233447 |    No |
|    173 |         Hannah |          Taylor |   hannah@gmail.com    |          hannaht |     0112233448 |    No |
|    174 |          Isaac |        Anderson |    isaac@gmail.com    |           isaaca |     0112233449 |    No |
|    175 |          Julia |          Thomas |     juli@gmail.com    |           juliat |     0112233450 |    No |
|    188 |         Albert |           Smith |   albert@gmail.com    |          alberts |     0112233441 |    No |
|    189 |            mna |             kla |      lala@gmail.com   |             lala |   12121212121 |    No |
|    190 |       Alwaleed |          Hassan |   welley@gmail.com    |           Welly |   60014908386 |    No |
+--------+-----------------+-----------------+-----------------------+-----------------+-----------------+-------+

Enter the User ID to delete or 0 to return: 189

Are you sure you want to delete User ID 189? (y/n): y

User ID 189 deleted successfully!
```

Figure 3.45: Delete users page

~PAYMANT MANAGEMENT~

```
*****************************************************************
*                        PAYMENT MANAGEMENT                     *
*****************************************************************

1. View All Payments
2. Add New Payment
3. Update Payment Information
4. Delete Payment
5. Return to Admin Page

Enter your choice: |
```

Figure 3.46: Payment management page

```
*************************************************************
*                     VIEW ALL PAYMENTS                     *
*************************************************************

Options:
1. View All Payments
2. Search Payment by Payment ID
3. Return to Main Menu
Enter your choice: 2

Enter Payment ID to search: 2

+-------------+-------------+-------------+------------------+------------------+---------------------+-----------------+
| Payment ID  | User ID     | Booking ID  | Payment Method   | Payment Amount   | Payment Date        | Payment Confirmed |
+-------------+-------------+-------------+------------------+------------------+---------------------+-----------------+
|           2 |         152 |           2 |       MasterCard |            40.00 | 2021-06-20 10:14:52 |             Yes |
+-------------+-------------+-------------+------------------+------------------+---------------------+-----------------+
```

```
*************************************************************
*                  VIEW ALL PAYMENTS - PAGE                 *
*************************************************************

+-------------+-------------+-------------+------------------+------------------+---------------------+-----------------+
| Payment ID  | User ID     | Booking ID  | Payment Method   | Payment Amount   | Payment Date        | Payment Confirmed |
+-------------+-------------+-------------+------------------+------------------+---------------------+-----------------+
|           1 |         151 |           1 |             Visa |            25.00 | 2021-05-15 00:12:02 |             Yes |
|           2 |         152 |           2 |       MasterCard |            40.00 | 2021-06-20 10:14:52 |             Yes |
|           3 |         153 |           3 |             Cash |            15.00 | 2021-07-01 11:16:16 |             Yes |
|           4 |         154 |           4 |             Visa |            45.00 | 2022-01-12 08:15:13 |             Yes |
|           5 |         155 |           5 |       MasterCard |            30.00 | 2022-03-10 13:13:26 |             Yes |
|           6 |         156 |           6 |             Cash |            20.00 | 2022-06-15 06:00:32 |             Yes |
|           7 |         157 |           7 |             Visa |            20.00 | 2022-08-19 04:35:00 |             Yes |
|           8 |         158 |           8 |       MasterCard |            12.50 | 2022-09-25 09:30:21 |             Yes |
|           9 |         159 |           9 |             Cash |            35.00 | 2023-01-05 08:24:20 |             Yes |
|          10 |         160 |          10 |             Visa |            40.00 | 2023-03-18 04:10:38 |             Yes |
+-------------+-------------+-------------+------------------+------------------+---------------------+-----------------+

Navigation:
2. Next Page
3. Return to Main Menu
Enter your choice: |
```

Figure 3.47: View all payment & search

```
*****************************************************************
*                        ADD NEW PAYMENT                        *
*****************************************************************
Note: Enter '0' at any input prompt to return to the previous page.

Enter User ID: 2
Enter Booking ID: 211
Enter Payment Method (Visa/MasterCard/Cash): Visa
Enter Payment Amount: 22
Enter Payment Date (YYYY-MM-DD): 2025-12-19

Payment added successfully!

Would you like to add another payment? (y/n): |
```

Figure 3.48: Add new payment page

```
*************************************************************
*                    UPDATE USER INFORMATION                *
*************************************************************

+-------+----------------+----------------+-------------------+-------------------+-----------------+-------+-----------+-------------------+
| UserID | First Name     | Last Name      | Email             | Username          | Phone Number    | Admin | IsStudent | Matric Number     |
+-------+----------------+----------------+-------------------+-------------------+-----------------+-------+-----------+-------------------+
| 145    | Hassan         | Elyas          | alw@gmail.com     | hassan22          | 09128373632     | No    | Yes       | b0323107622       |
| 146    | Abdulah        | khaled         | abdu22@gmail.com  | abdu220           | 1234566789      | No    | Yes       | B032310752        |
| 147    | Ahmed          | naseer         | hamody@gmail.com  | hamody12          | 0981276665      | No    | Yes       | b21334566         |
| 148    | Lily           | Walker         | lily@gmail.com    | lilyw             | 0123456778      | No    | No        | N/A               |
| 149    | Nina           | Evans          | nina@gmail.com    | ninae             | 0123456776      | No    | No        | N/A               |
| 150    | Alice          | Johnson        | alice@gmail.com   | alicej            | 0123456789      | No    | No        | N/A               |
| 151    | Bob            | Smith          | bob@gmail.com     | bobsmith          | 0123456788      | No    | No        | N/A               |
| 152    | Charlie        | Brown          | charlie@gmail.com | charlieb          | 0123456787      | No    | No        | N/A               |
| 153    | Daisy          | Taylor         | daisy@gmail.com   | daisyt            | 0123456786      | No    | Yes       | M123456           |
| 154    | Evan           | Williams       | evan@gmail.com    | evanw             | 0123456785      | No    | No        | N/A               |
+-------+----------------+----------------+-------------------+-------------------+-----------------+-------+-----------+-------------------+

Options:
1. Previous Page
2. Next Page
3. Enter User ID to Update
4. Return to User Management
Enter your choice: 3

Enter the User ID to update: 2

Options:
1. First Name
2. Last Name
3. Phone Number
4. Email
5. Username
6. Password
7. Admin (1 for Yes, 0 for No)
8. Is Student (1 for Yes, 0 for No)
9. Matric Number
10. Back to User List

Enter your choice: |
```

Figure 3.49: Update payment page

```
*************************************************************
*                       DELETE USER                         *
*************************************************************

+-------+----------------+----------------+-------------------+-------------------+-----------------+-------+
| UserID| First Name     | Last Name      | Email             | Username          | Phone Number    | Admin |
+-------+----------------+----------------+-------------------+-------------------+-----------------+-------+
| 2     | asjed          | tomaro         | aj@gmail.com      | asjed22           | 0908112121      | No    |
| 145   | Hassan         | Elyas          | alw@gmail.com     | hassan22          | 09128373632     | No    |
| 146   | Abdulah        | khaled         | abdu22@gmail.com  | abdu220           | 1234566789      | No    |
| 147   | Ahmed          | naseer         | hamody@gmail.com  | hamody12          | 0981276665      | No    |
| 148   | Lily           | Walker         | lily@gmail.com    | lilyw             | 0123456778      | No    |
| 149   | Nina           | Evans          | nina@gmail.com    | ninae             | 0123456776      | No    |
| 150   | Alice          | Johnson        | alice@gmail.com   | alicej            | 0123456789      | No    |
| 151   | Bob            | Smith          | bob@gmail.com     | bobsmith          | 0123456788      | No    |
| 152   | Charlie        | Brown          | charlie@gmail.com | charlieb          | 0123456787      | No    |
| 153   | Daisy          | Taylor         | daisy@gmail.com   | daisyt            | 0123456786      | No    |
| 154   | Evan           | Williams       | evan@gmail.com    | evanw             | 0123456785      | No    |
| 155   | Fiona          | Clark          | fiona@gmail.com   | fionac            | 0123456784      | No    |
| 156   | George         | Miller         | george@gmail.com  | georgem           | 0123456783      | No    |
| 157   | Hannah         | Davis          | nah@gmail.com     | hannahd           | 0123456782      | No    |
| 158   | Ian            | Moore          | ian@gmail.com     | ianm              | 0123456781      | No    |
| 159   | Julia          | Anderson       | julia@gmail.com   | juliaa            | 0123456780      | No    |
| 160   | Kevin          | Harris         | kevin@gmail.com   | kevinh            | 0123456779      | No    |
| 162   | Mike           | Scott          | mike@gmail.com    | mikesc            | 0123456777      | No    |
| 164   | Oscar          | Wilson         | oscar@gmail.com   | oscarw            | 0123456775      | No    |
| 165   | Paula          | Reed           | paula@gmail.com   | paular            | 0123456774      | No    |
| 166   | Alwaleed       | Hassan         | welly@gmail.com   | alwaleed24        | 60149408286     | Yes   |
| 167   | Brianna        | Johnson        | brianna@gmail.com | briannaj          | 0112233442      | No    |
| 168   | Christopher    | Brown          | chris@gmail.com   | chrisb            | 0112233443      | No    |
| 169   | Diana          | Davis          | diana@gmail.com   | dianad            | 0112233444      | No    |
| 170   | Ethan          | Miller         | ethan@gmail.com   | ethanm            | 0112233445      | No    |
| 171   | Fiona          | Wilson         | fiofo@gmail.com   | fionaw            | 0112233446      | No    |
| 172   | George         | Moore          | gerge@gmail.com   | gergem            | 0112233447      | No    |
| 173   | Hannah         | Taylor         | hannah@gmail.com  | hannaht           | 0112233448      | No    |
| 174   | Isaac          | Anderson       | isaac@gmail.com   | isaaca            | 0112233449      | No    |
| 175   | Julia          | Thomas         | juli@gmail.com    | juliat            | 0112233450      | No    |
| 188   | Albert         | Smith          | albert@gmail.com  | alberts           | 0112233441      | No    |
| 190   | Alwaleed       | Hassan         | welley@gmail.com  | Welly             | 60014908386     | No    |
| 195   | dan            | lan            | dan@gmail.com     | dan11             | 60123233432     | No    |
| 196   | fal            | kla            | fal@gmail.com     | fa11              | 60124567891     | No    |
+-------+----------------+----------------+-------------------+-------------------+-----------------+-------+

Enter the User ID to delete or 0 to return: 196

Are you sure you want to delete User ID 196? (y/n): y

User ID 196 deleted successfully!
```

Figure 3.50: Delete payment page

~MEALS MANAGEMNET~

```
*************************************************************
*                      MEALS MANAGEMENT                     *
*************************************************************
1. View Meals
2. Add Meal
3. Update Meal
4. Return to Admin Page
*************************************************************
Enter your choice: |
```

Figure 3.51: Meals management

```
*************************************************************
*                        VIEW MEALS                         *
*************************************************************

Options:
1. View All Meals
2. Search Meal by Name
3. Return to Main Menu
Enter your choice: 1

+-------+-------------------------+-------------------+
| MealID| Meal Name               | Meal Price (RM)   |
+-------+-------------------------+-------------------+
|     1 |             Popcorn (S) |              5.00 |
|     2 |             Popcorn (M) |              7.50 |
|     3 |             Popcorn (L) |             10.00 |
|     4 |          Soft Drink (S) |              3.00 |
|     5 |          Soft Drink (M) |              5.00 |
|     6 |          Soft Drink (L) |              6.50 |
|     7 |                  cocola |             10.50 |
|     8 |                   vanta |             14.00 |
|     9 |       Nachos with Cheese |              8.00 |
|    10 |                  Hot Dog |              6.50 |
|    11 |          Chicken Nuggets |              7.00 |
|    12 |            Fries (Small) |              4.00 |
|    13 |            Fries (Large) |              6.00 |
|    14 |           Ice Cream Cup |              4.50 |
|    15 |            Water Bottle |              2.00 |
|    22 |                 chocata |             21.00 |
|   111 |                   Frise |              9.00 |
+-------+-------------------------+-------------------+
```

```
*************************************************************
*                        VIEW MEALS                         *
*************************************************************

Options:
1. View All Meals
2. Search Meal by Name
3. Return to Main Menu
Enter your choice: 2

Enter Meal Name to search: PO

+-------+-------------------------+-------------------+
| MealID| Meal Name               | Meal Price (RM)   |
+-------+-------------------------+-------------------+
|     1 |             Popcorn (S) |              5.00 |
|     2 |             Popcorn (M) |              7.50 |
|     3 |             Popcorn (L) |             10.00 |
+-------+-------------------------+-------------------+
```

Figure 3.52: View all meals & search

```
*************************************************************
*                       ADD NEW MEAL                        *
*************************************************************

Enter Meal ID: 121
Enter Meal Name: TEA
Enter Meal Price (RM): 22

Meal added successfully!
```

Figure 3.53: Add meals page

```
*************************************************************
*                       UPDATE MEAL                         *
*************************************************************

+-------+-------------------------+-------------------+
| MealID| Meal Name               | Meal Price (RM)   |
+-------+-------------------------+-------------------+
|     1 |             Popcorn (S) |              5.00 |
|     2 |             Popcorn (M) |              7.50 |
|     3 |             Popcorn (L) |             10.00 |
|     4 |          Soft Drink (S) |              3.00 |
|     5 |          Soft Drink (M) |              5.00 |
|     6 |          Soft Drink (L) |              6.50 |
|     7 |                  cocola |             10.50 |
|     8 |                   vanta |             14.00 |
|     9 |       Nachos with Cheese |              8.00 |
|    10 |                  Hot Dog |              6.50 |
|    11 |          Chicken Nuggets |              7.00 |
|    12 |            Fries (Small) |              4.00 |
|    13 |            Fries (Large) |              6.00 |
|    14 |           Ice Cream Cup |              4.50 |
|    15 |            Water Bottle |              2.00 |
|    22 |                 chocata |             21.00 |
|   111 |                   Frise |              9.00 |
|   121 |                     TEA |             22.00 |
+-------+-------------------------+-------------------+

Enter Meal ID to update: 121
Enter new Meal Name: COOFE
Enter new Meal Price (RM): 9

Meal updated successfully!
```

Figure 3.54: Update meals

~MOVIES MANAGMENET~

```
*****************************************************************
*                        MOVIE MANAGEMENT                       *
*****************************************************************
1. Add Movie
2. Delete Movie
3. View Movies
4. Update Movie
5. Return to Admin Page
*****************************************************************
Enter your choice:
```

Figure 3.55: Movies management

```
*************************************************************
*                        VIEW ALL MOVIES                    *
*************************************************************

Options:
1. View All Movies
2. Search Movie
3. Return to Main Menu
Enter your choice: 2

Enter Movie Title to search: FRO

+-------+---------------------------------+---------------+------------+--------+----------+
| ID    | Title                           | Genre         | ReleaseDate| Rating | Duration |
+-------+---------------------------------+---------------+------------+--------+----------+
|    14 |                       Frozen III |     Animation | 2025-06-10 |    8.5 |     1:55 |
+-------+---------------------------------+---------------+------------+--------+----------+
```

```
*************************************************************
*                        VIEW ALL MOVIES                    *
*************************************************************

Options:
1. View All Movies
2. Search Movie
3. Return to Main Menu
Enter your choice: 2

Enter Movie Title to search: jo

+-------+---------------------------------+---------------+------------+--------+----------+
| ID    | Title                           | Genre         | ReleaseDate| Rating | Duration |
+-------+---------------------------------+---------------+------------+--------+----------+
|    13 |             John Wick: Chapter 5 |        Action | 2025-03-15 |    9.2 |     2:45 |
|    18 |                   Indiana Jones |     Adventure | 2023-06-30 |    7.5 |     2:22 |
|    45 |                           Joker |         drama |   00-00-00 |    9.4 |     3:35 |
+-------+---------------------------------+---------------+------------+--------+----------+
```

Figure 3.56: View movies & search page

```
*************************************************************
*                          ADD MOVIE                        *
*************************************************************

Enter Movie Title: Joker
Enter Movie Genre: drama
Enter Release Date (YYYY-MM-DD): 2019-1200
Enter Rating (0.0 to 10.0): 9.4
Enter Duration (e.g., 2h 15m): 3:35

Movie added successfully!
```

Figure 3.57: Add movies page

```
*****************************************************************
*                          DELETE MOVIE                         *
*****************************************************************

+----------+----------------------------------------+-----------------+--------------+--------+----------+
| Movie ID | Title                                  | Genre           | Release Date | Rating | Duration |
+----------+----------------------------------------+-----------------+--------------+--------+----------+
| 11       | Aquaman                                | Action          | 2024-12-25   | 7.9    | 2:30     |
| 12       | Wonka                                  | Adventure       | 2024-12-13   | 8.0    | 2:10     |
| 13       | John Wick: Chapter 5                   | Action          | 2025-03-15   | 9.2    | 2:45     |
| 14       | Frozen III                             | Animation       | 2025-06-10   | 8.5    | 1:55     |
| 15       | Avengers: Secret Wars                  | Action          | 2025-11-08   | 9.3    | 2:50     |
| 16       | The Batman: Part II                    | Action          | 2025-03-04   | 9.0    | 2:55     |
| 17       | Shazam! Fury                           | Fantasy         | 2023-03-17   | 7.3    | 2:10     |
| 18       | Indiana Jones                          | Adventure       | 2023-06-30   | 7.5    | 2:22     |
| 19       | Fast X                                 | Action          | 2023-05-19   | 7.2    | 2:20     |
| 20       | Super Mario Bros                       | Animation       | 2023-04-07   | 7.8    | 1:32     |
+----------+----------------------------------------+-----------------+--------------+--------+----------+

Options:
1. Previous Page
2. Next Page
3. Enter Movie ID to Delete
4. Return to Main Menu
Enter your choice: 3

Enter the Movie ID of the movie to delete: 45

Movie deleted successfully!
```

Figure 3.58: Delete movies page

```
What would you like to update?
1. Title
2. Genre
3. Release Date
4. Rating
5. Duration
6. Back to Movie List
Enter your choice: |
```

Figure 3.59: Update movies page

~BOOKING MANAGEMNET~

```
*****************************************************************
*                        MANAGE BOOKINGS                        *
*****************************************************************
1. View All Bookings
2. Add Booking
3. Update Booking
4. Return to Admin Page

Enter your choice: |
```

Figure 3.60: Booking management

```
*************************************************************
*                     VIEW ALL BOOKINGS                     *
*************************************************************

Options:
1. View All Bookings
2. Search Booking by ID
3. Return to Main Menu
Enter your choice: 2

Enter Booking ID to search: 3

+-----------+----------+----------+-------------+------------+---------+-----------------+---------------+------------+
| BookingID | Movie_id | User_id  | Screen Type | BookingDate| TimeSlot| Seats Booked    | Ticket Types  | Total Price|
+-----------+----------+----------+-------------+------------+---------+-----------------+---------------+------------+
|         3 |        3 |      153 |          3D | 2021-07-01 |   16:00 |              C1 |       Student |      15.00 |
+-----------+----------+----------+-------------+------------+---------+-----------------+---------------+------------+
```

```
*************************************************************
*                  VIEW ALL BOOKINGS - PAGE                 *
*************************************************************

+-----------+----------+----------+-------------+------------+---------+-----------------+---------------+------------+
| BookingID | Movie_id | User_id  | Screen Type | BookingDate| TimeSlot| Seats Booked    | Ticket Types  | Total Price|
+-----------+----------+----------+-------------+------------+---------+-----------------+---------------+------------+
|        11 |       11 |      148 |        IMAX | 2023-05-22 |   18:00 |              D1 |    Individual |      22.00 |
|        12 |       12 |      162 |    Standard | 2023-07-08 |   16:00 |              B1 |    Individual |      30.00 |
|        13 |       13 |      149 |          3D | 2023-08-15 |   19:00 |              A1 |          kids |      10.00 |
|        14 |       14 |      164 |         VIP | 2023-09-20 |   21:00 |           C1,C2 |   kids,Senior |      50.00 |
|        15 |       15 |      151 |    Standard | 2023-10-12 |   20:00 |           H1,H2 |   Senior,kids |      28.00 |
|        76 |       12 |      149 |    Standard | 2023-01-10 |  10:00  |              D3 |    Individual |      27.00 |
|        77 |       11 |      148 |    Standard | 2023-01-10 |  10:00  |           D2,D3 |    Individual |      66.50 |
|        78 |       11 |      145 |    Standard | 2025-01-10 |  10:00  |              G6 |       Student |      27.50 |
|        80 |       12 |      148 |    Standard | 2025-01-09 |  01:00  |           D2,D3 |       Student |      42.50 |
|        82 |       12 |      148 |         VIP | 2025-01-10 |  01:00  |              D2 |    Individual |      39.50 |
+-----------+----------+----------+-------------+------------+---------+-----------------+---------------+------------+

Navigation:
1. Previous Page
2. Next Page
3. Return to Main Menu
Enter your choice: |
```

Figure 3.61: View booking & search

```
*****************************************************************
*                        ADD BOOKING                            *
*****************************************************************
Note: Enter '0' at any input prompt to return to the previous page.


+-----------+----------+----------+-------------+-------------+---------+-----------------+---------------+-------------+
| BookingID | Movie ID | User ID  | Screen Type | BookingDate| TimeSlot| Seats Booked    | Ticket Types  | Total Price|
+-----------+----------+----------+-------------+-------------+---------+-----------------+---------------+-------------+
|         1 |        1 |      151 |    Standard | 2021-05-15 |   18:00 |              A1 |    Individual |       25.00 |
|         2 |        2 |      152 |        IMAX | 2021-06-20 |   20:00 |           B3,B4 | Individual, kids |      40.00 |
|         3 |        3 |      153 |          3D | 2021-07-01 |   16:00 |              C1 |       Student |       15.00 |
|         4 |        4 |      154 |         VIP | 2022-01-12 |   19:00 |           D1,D2 | Individual, kids, |      45.00 |
|         5 |        5 |      155 |    Standard | 2022-03-10 |   15:00 |           E1,E2 | Senior, Individual |      30.00 |
|         6 |        6 |      156 |        IMAX | 2022-06-15 |   21:00 |              F1 |         Adult |       20.00 |
|         7 |        7 |      157 |          3D | 2022-08-19 |   17:00 |           G1,G2 |    kids, kids |       20.00 |
|         8 |        8 |      158 |    Standard | 2022-09-25 |   19:00 |              H1 |    Individual |       12.50 |
|         9 |        9 |      159 |         VIP | 2023-01-05 |   20:00 |           C1,C2 | Senior,Individual |      35.00 |
|        10 |       10 |      160 |    Standard | 2023-03-18 |   14:00 |           B2,B1 | Individual, kids |      40.00 |
+-----------+----------+----------+-------------+-------------+---------+-----------------+---------------+-------------+

Options:
2. Next Page
3. Add New Booking
4. Return to Booking Management
Enter your choice: 3

Enter Movie ID: 2
Enter User ID: 2
Enter Screen Type: IMAX
Enter Booking Date (YYYY-MM-DD): 2025-01-16
Enter Time Slot: 10:00
Enter Seats Booked (comma-separated): B2
Enter Ticket Types: kids
Enter Total Price: 15

Booking added successfully!

Would you like to add another booking? (y/n): |
```

Figure 3.62: Add booking page

```
Options to Update:
1. Movie ID
2. User ID
3. Screen Type
4. Booking Date
5. Time Slot
6. Seats Booked
7. Ticket Types
8. Total Price
9. Back to Booking List
Enter your choice: |
```

Figure 3.63: Update booking page

~ REPORT MANAGEMNET ~

```
*************************************************************
*                          MANAGE REPORT                    *
*************************************************************

Please choose an option:
1. View Full Report
2. Profit Margin Analysis
3. Graph Sales Summary
4. Bookings by Destination
0. Exit
*************************************************************
Enter your choice:
```

Figure 3.64: **Report management**

```
*************************************************************
*                        SALES REPORT                       *
*************************************************************

Available Booking Data Range: 2021-05-15 to 2025-01-21
*************************************************************

All Years and Months Sales Report:
*************************************************************
| Year      | Month     | Total Sales (RM)                 |
*************************************************************
| 2021      | 5         | RM 25.00                         |
| 2021      | 6         | RM 40.00                         |
| 2021      | 7         | RM 15.00                         |
| 2021      | 8         | RM 65.00                         |
| 2021      | 9         | RM 80.00                         |
| 2021      | 10        | RM 70.00                         |
| 2021      | 11        | RM 102.00                        |
| 2021      | 12        | RM 110.00                        |
| 2022      | 1         | RM 45.00                         |
| 2022      | 2         | RM 10.00                         |
| 2022      | 3         | RM 45.00                         |
| 2022      | 4         | RM 160.00                        |
| 2022      | 5         | RM 9.00                          |
| 2022      | 6         | RM 210.00                        |
| 2022      | 7         | RM 80.00                         |
| 2022      | 8         | RM 30.00                         |
| 2022      | 9         | RM 37.50                         |
| 2022      | 10        | RM 155.00                        |
| 2022      | 11        | RM 7.00                          |
| 2022      | 12        | RM 500.00                        |
| 2023      | 1         | RM 128.50                        |
| 2023      | 2         | RM 90.00                         |
| 2023      | 3         | RM 40.00                         |
| 2023      | 4         | RM 135.00                        |
| 2023      | 5         | RM 22.00                         |
| 2023      | 6         | RM 105.00                        |
| 2023      | 7         | RM 85.00                         |
| 2023      | 8         | RM 10.00                         |
| 2023      | 9         | RM 50.00                         |
| 2023      | 10        | RM 28.00                         |
| 2023      | 11        | RM 105.00                        |
| 2023      | 12        | RM 110.00                        |
| 2024      | 1         | RM 95.00                         |
| 2024      | 2         | RM 10.00                         |
| 2024      | 3         | RM 70.00                         |
| 2024      | 4         | RM 90.00                         |
| 2024      | 5         | RM 20.00                         |
| 2024      | 6         | RM 24.00                         |
| 2024      | 7         | RM 40.00                         |
| 2024      | 8         | RM 45.00                         |
| 2024      | 9         | RM 35.00                         |
| 2024      | 10        | RM 60.00                         |
| 2024      | 11        | RM 12.00                         |
| 2024      | 12        | RM 17.00                         |
| 2025      | 1         | RM 891.00                        |
*************************************************************

1.Total Monthly Sales: RM 4113.00
  2.Total Inventory Value: RM 2879.10
  3.Profit Margin: 30.00%
*************************************************************

Press any key to return to the report menu...
```

Figure 3.65: **View report all of years**

```
************************************************************
*                    GRAPHICAL SALES SUMMARY               *
************************************************************

Sales Summary by Year:
************************************************************
2021 :   (RM 507.00)
2022 :    (RM 1288.50)
2023 :   (RM 908.50)
2024 :   (RM 518.00)
2025 :   (RM 891.00)
************************************************************

Enter a year to view monthly trends or 0 to go back: |
```

Figure 3.66: **View graph sales summary by year**

```
Monthly Sales for Year: 2022
************************************************************
January : *  (RM 45.00)
February : *  (RM 10.00)
March : **  (RM 45.00)
April : ***  (RM 160.00)
May : *  (RM 9.00)
June : **  (RM 210.00)
July : ***  (RM 80.00)
August : **  (RM 30.00)
September : **  (RM 37.50)
October : ****  (RM 155.00)
November : *  (RM 7.00)
December : *  (RM 500.00)
************************************************************

for example (1=January, 5=May...etc
Enter a month (1-12) to view detailed bookings or 0 to go back: |
```

Figure 3.67: **View graph sales summary by month**

```
Detailed Bookings for October 2022
*********************************************************************************
| Booking ID | User Name       | Movie Name       | Booking Date  |   Price      |
*********************************************************************************
| 163        | Nina Evans      | Super Mario Bros| 2022-10-01   | RM 30.00      |
| 164        | Alice Johnson   | Elemental       | 2022-10-10   | RM 40.00      |
| 165        | Bob Smith       | Spider Man      | 2022-10-20   | RM 60.00      |
| 166        | Charlie Brown   | The Flash       | 2022-10-25   | RM 25.00      |
*********************************************************************************

Press any key to return to the monthly summary...
```

Figure 3.68: **View graph sales summary: booking details for month**

```
*************************************************************
*                      BOOKINGS BY YEAR                     *
*************************************************************

Available Booking Years:
*************************************************************
2021       | (18 bookings)
2022       | (23 bookings)
2023       | (28 bookings)
2024       | (20 bookings)
2025       | (23 bookings)
*************************************************************

Enter a year to view monthly trends or 0 to go back:
```

Figure 3.69: Bookings report by year

```
Available Booking Months for Year: 2023
*************************************************************
January    | *** (3 bookings)
February   | *** (3 bookings)
March      | * (1 bookings)
April      | *** (3 bookings)
May        | * (1 bookings)
June       | *** (3 bookings)
July       | ***** (5 bookings)
August     | * (1 bookings)
September  | * (1 bookings)
October    | * (1 bookings)
November   | *** (3 bookings)
December   | *** (3 bookings)
*************************************************************

for example (1=January, 5=May...etc
Enter the month to view movie bookings or 0 to go back: |
```

Figure 3.70: Bookings report by month

```
Detailed Bookings for July 2023
*********************************************************************************
| Booking ID | User Name       | Movie Name       | Booking Date  |   Price      |
*********************************************************************************
| 12         | Mike Scott      | Wonka            | 2023-07-08    | RM 30.00     |
| 107        | Hassan Elyas    | Barbie           | 2023-07-17    | RM 24.00     |
| 105        | Hassan Elyas    | Oppenheimer      | 2023-07-21    | RM 9.00      |
| 104        | Lily Walker     | Oppenheimer      | 2023-07-21    | RM 15.00     |
| 109        | Alice Johnson   | Oppenheimer      | 2023-07-22    | RM 7.00      |
*********************************************************************************

Press any key to return to the report menu...
```

Figure 3.71: Report of movie booking for month

```
*******************************************************************
*                     PROFIT MARGIN ANALYSIS                      *
*******************************************************************

Enter the range for analysis:
Start Date (YYYY-MM): 2023-07
End Date (YYYY-MM): 2024-01

Profit Margin Analysis:
*******************************************************************
|     Period Range     |   Total Profit (RM)   |     Change (%)      |
*******************************************************************
| 2023-07 to 2023-08| RM 3.00                |                  -88.24% |
| 2023-08 to 2023-09| RM 15.00               | +                400.00% |
| 2023-09 to 2023-10| RM 8.40                |                  -44.00% |
| 2023-10 to 2023-11| RM 31.50               | +                275.00% |
| 2023-11 to 2023-12| RM 33.00               | +                4.76% |
| 2023-12 to 2024-01| RM 28.50               |                  -13.64% |
*******************************************************************
```

Figure 3.72: Profit margin analysis

CHAPTER 4: IMPLEMENTATION

4.1 Naming Convention

Some examples:

```
// Admin page
void adminPage();

// payment Management
void managePayments();
void viewAllPayments();
void addNewPayment();
void updatePayment();
void deletePayment();

// movie Management
void manageMovies();
void addMovie();
void deleteMovie();
void viewAllMovies();
void updateMovie();

// users Management
void manageUsers();
void viewAllUsers();
void addNewUser();
void updateUser();
void deleteUser();
```

In this section, we see a well structured breakdown of admin functionalities. Each category, such as **Payments, Movies, Users**, has its own management functions. This approach improves readability and maintainability. Whenever an admin needs to perform an action, they can easily find the corresponding function without unnecessary complexity.

```
void displayMovies(const string& searchKeyword = "");

void browseAndSearchMovies(int userID, bool isStudent);

void displaySearchMoviesPage();

void displayDaysPage(const std::string& chosenMovie, int userID, int movieID, bool isStudent);
void displayScreenAndTimePage(int userID, int movieID, const string& chosenDay, bool isStudent);

void displaySeatsPage(const std::string& chosenDay, const std::string& chosenTimeSlot, const std::string& screenType, int userID, int movieID, bool allowSelection);
```

The separation of **displaying** and **searching** ensures the system is efficient and organized. Instead of one massive function that does everything, each aspect of the user journey (browsing, selecting, booking) is modular

4.2 Function

```
double calculateTotalPrice(const vector<string>& ticketTypes, bool isStudent = false) {
    map<string, double> ticketPrices = {
        {"Individual", 22.0},
        {"Kids", 15.0},
        {"Senior", 18.0},
        {"Student", 15.0}
    };

    double total = 0.0;

    for (const auto& ticket : ticketTypes) {
        if (ticketPrices.find(ticket) != ticketPrices.end()) {
            total += ticketPrices[ticket];
        }
    }

    return total;
}
```

This function follows modular programming in that it has kept the price calculation separate so that it could be reused for other parts of the system. It ensures data accuracy through the use of conditional logic that differentiates pricing between students and normal users.

4.3 Array

```
int rows = 8, cols = 8;
vector<vector<string>> seats(rows, vector<string>(cols));
char rowLetter = 'A';

for (int i = 0; i < rows; ++i) {
    for (int j = 0; j < cols; ++j) {
        seats[i][j] = rowLetter + to_string(j + 1);
    }
    rowLetter++;
}
```

In the example above, we used a 2D vector (vector<vector<string>>), which is a dynamic array to store the seat arrangement in a cinema hall. Each element in this 2D vector represents a seat label (e.g., "A1", "B2"), where the rows correspond to seat rows (A-H) and columns correspond to seat numbers (1-8). This structure allows for flexible memory allocation, making it easy to modify the seat arrangement dynamically.

4.4 Selection

```
if (seatChoice == "done") {
    if (selectedSeats.empty()) {
        cout << "\033[41;97mPlease select at least one seat.\033[0m\n";
        system("pause");
        continue;
    }
    break;
```

In the Seat Selection Page function, selection statements play a crucial role in validating user input, handling errors, and controlling navigation.

4.5 Control

```
string input;
int choice = 0;

do {
    system("cls");
    cout << "\n\n\n\n\n\n\n\n\n\n";
    cout << "                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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    getline(cin, input);

    if (input.length() == 1 && isdigit(input[0])) {
        choice = stoi(input);
    }
    else {
        choice = 0;
    }

    switch (choice) {
    case 1:
        system("cls");
        registerUserPage();
        break;
    case 2:
        system("cls");
        loginUserPage();
        break;
    case 3:
        cout << "Exiting the system. Goodbye!\n";
        break;
    default:
        setConsoleColor(12);
        cout << "\033[31mInvalid choice. Please choose 1, 2, or 3.\033[0m\n";
        setConsoleColor(7);
        system("pause");
    }
} while (choice != 3);
```

```
**************************************\n";
*       CINEMA TICKET BOOKING SYSTEM    *\n";
**************************************\n\n";
1. REGISTER\n";
2. LOGIN\n";
3. EXIT\n\n";
Choose an option: ";
```

The combination of loops, conditions, and switch cases effectively controls the user's interaction with the system, ensuring smooth navigation while preventing invalid inputs from causing errors.

4.6 Pointer

```
void registerUser(const string& firstName, const string& lastName, const string& phoneNumber,
    const string& email, const string& userName, const string& password,
    const string& date, const string& matricNumber, bool isStudent) {
    try {
        sql::PreparedStatement* pstmt = globalCon->prepareStatement(
            "INSERT INTO users (FirstName, LastName, PhoneNumber, Email, UserName, Password, Date, MatricNumber, IsStudent) "
            "VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?)");

        pstmt->setString(1, firstName);
        pstmt->setString(2, lastName);
        pstmt->setString(3, phoneNumber);
        pstmt->setString(4, email);
        pstmt->setString(5, userName);
        pstmt->setString(6, password);
        pstmt->setString(7, date);
        pstmt->setString(8, matricNumber.empty() ? "N/A" : matricNumber);
        pstmt->setBoolean(9, isStudent);

        pstmt->executeUpdate();
        cout << "User registered successfully!" << endl;

        delete pstmt;
    }
    catch (sql::SQLException& e) {
        cerr << "SQL Error: " << e.what() << " (MySQL error code: " << e.getErrorCode()
            << ", SQLState: " << e.getSQLState() << ")" << endl;
    }
}
```

In the given ()registerUser function, pointers are used to dynamically allocate a prepared statement for executing SQL queries.

4.7 Error Handling

```
try {
    // Fetch the current value from the database
    sql::PreparedStatement* fetchStmt = globalCon->prepareStatement(
        "SELECT " + fieldName + " FROM users WHERE User_id = ?");
    fetchStmt->setInt(1, userID);
    sql::ResultSet* res = fetchStmt->executeQuery();

    if (res->next()) {
        currentValue = res->getString(fieldName);
    }
    delete res;
    delete fetchStmt;
}
catch (sql::SQLException& e) {
    cerr << "SQL Error (fetch current value): " << e.what() << endl;
    system("pause");
    continue;
}
```

In the given ()updateUserProfile function, the system checks if the user input is valid before proceeding. If the input is out of range, missing, or does not meet the required format, the system will display an error message and ask the user to reenter the correct information.

CHAPTER 5: CONCLUSION

5.1 Constraints

Building the Cinema Ticket Booking System comes with a few challenges that shape its overall design and usability. One of the key limitations is the use of a text-based interface, which restricts the ability to include modern graphical features, making it less visually appealing compared to web based platforms. Another challenge lies in ensuring the database can handle a large number of bookings, seat allocations, and user transactions seamlessly, while also preventing issues like duplicate reservations.

Security is another significant concern. Protecting the system against risks like SQL injection and unauthorized access means implementing strong authentication measures, which adds to the technical workload. Time is also a pressing factor creating a fully functional and user-friendly system within a tight deadline requires careful prioritization and efficient execution. Furthermore, the system depends on external resources like database servers and Internet connectivity, making it vulnerable to potential disruptions. To address this, planning for offline access and server backups is essential.

By tackling these constraints thoughtfully, we aim to deliver a robust, easy to use cinema ticketing system that satisfies the needs of both users and administrators, ensuring a smooth and enjoyable booking experience.

5.2 Future Improvements

The Cinema Ticket Booking System can be extended further by overcoming the shortcomings of the present system. The most important upgrade would be changing the text-based interface to a graphical user interface, thus making the system more visually appealing and user-friendly. This can be done by developing a web based or mobile application version for seamless experiences across different devices.

One may enhance the database performance constraints by query optimization, implementation of indexing, and using cloud based database solutions. This will ensure a large volume of bookings are managed efficiently without duplicate reservations. Security features may be enhanced by the integration of multi factor authentication, encryption protocols, and stricter user access control to mitigate risks associated with SQL injection and unauthorized accesses.

Future versions could include real-time seat availability with no delays from the moment an update occurs and is available, integrating extra payment integrations using e-wallets, cryptocurrency, and online banking integrations that make life easier for its users. Added support for offline access and mechanisms for automated redundancy of servers to ensure continued system availability despite connectivity issues also improves system design.

The Cinema Ticket Booking System will be powered up with these enhancements to be efficient, modernized, and secure to meet changing needs for both customers and administrators.

REFERENCES

- Smith, J. A., & Brown, L. (2015). *Database Management Systems: Design and Implementation*. New York: McGraw-Hill Education.
- Zhang, H., & Wang, X. (2018). Efficient data retrieval techniques in modern relational databases. *Journal of Computer Science*, 45(3), 102–114.
- Johnson, K. R., & Tan, Y. M. (2013). *Web Application Development: From Concept to Deployment*. London: Wiley.
- IEEE Standard 829-2008, *IEEE Standard for Software Test Documentation*. IEEE, 2008.
- Kumar, R., & Gupta, M. (2019). Advances in SQL query optimization techniques: A review. *International Journal of Computer Science and Technology*, 10(4), 45–56.
- Wirth, N. (1996). *Algorithms + Data Structures = Programs*. Englewood Cliffs, N.J.: Prentice Hall.
- Stewart, D. W., & Usher, J. E. (2017). *Principles of User Interface Design: Practical Approaches for Interactive Systems*. Chicago: University Press.
- *Database Management Systems - An Overview*
  Accessed: 12 January 2025.
  https://www.geeksforgeeks.org/database-management-system/
- *Web Application Development Best Practices*
  Accessed: 14 January 2025.
  https://www.smashingmagazine.com/guides/web-applications/
- *Introduction to Software Testing*
  Accessed: 16 January 2025.
  https://www.softwaretestinghelp.com/software-testing-tutorials/
- TGV cinema